Credit Shocks and Populism*

Nicolò Fraccaroli†
Brown University

Alessandro Pizzigolotto‡
University of Copenhagen

This Version: September 2026

Abstract

This paper shows that credit shocks are key drivers of populist sentiment. Exploiting spatial variation in exposure to an exogenous lending cut by a large German bank in 2007–08, we find that exposure to the shock causes a persistent increase in populist political preferences. Using machine learning applied to German parliamentary speeches, we show that the shift favours parties combining populist rhetoric with attention to banking issues. County-level evidence suggests that the mechanism operates through voters' perceptions of local economic decline. A decomposition of individual causal effects highlights labour market history as the main factor shaping voters' populist reaction to the credit shock.

JEL Classification: P16; G21; D72; E51.

Keywords: Populism; Credit; Banking Crisis; Electoral Behaviour.

*We thank Jakob Beuschlein, Jeanet Sinding Bentzen, Michele Cantarella, Sirus Dehdari, Mitch Downey, Andreas Haller, Casper Worm Hansen, Kilian Huber, Andreas Kern, Ro'ee Levy, Sebastian Ottinger, Jesse Shapiro, Fabian Sinn and Matia Vannoni for their insightful comments and fruitful discussions. Alessandro is extremely grateful to Katrine Løken and Jósef Sigurdsson for their invaluable advice and precious guidance. We also thank the outstanding community at the Norwegian School of Economics (NHH) and Stockholm University, David Strömberg and all the participants in the Labour Group Supervision at Stockholm University for the many conversations that made this paper possible. The paper also benefited from feedback from seminar participants at NHH, Brown University, Institute of New Economic Thinking, Scandinavian PhD Seminar, University of Rome Tor Vergata, Stockholm University, and University of Gothenburg. We are also grateful for the feedback from participants at the following conferences: EPSA Annual Virtual Conference 2021, 2nd Monash-Warwick-Zurich Text-as-Data Workshop, the IFABS 2021 Oxford Conference, the 2021 Bank of England-European Central Bank-King's College Advanced Analytics Conference, the 2022 MPSA conference, the 5th Annual COMPTEXT Conference, the poster session at the FAIR Midway Conference 2022, and the Graduate Students in International Political Economy (GSIPE) 2023 Workshop.

†William R. Rhodes Center for International Economics and Finance. nicolo_fraccaroli@brown.edu.

‡Corresponding Author. University of Copenhagen, Department of Economics. alpi@econ.ku.dk.

1 Introduction

Following the Great Financial Crisis, populist parties achieved significant electoral success in a number of countries.[1] This pattern has led many observers to believe that the financial crises caused the rise of populism in affected countries.[2] However, despite the fact that the recession originated in a banking crisis characterised by a sharp contraction in lending (Ivashina and Scharfstein, 2010), existing empirical research on the economic causes of populism focused on other channels, such as unemployment or trade shocks. As a result, the direct effect of bank lending on support for populist parties remains largely unexplored.

This paper investigates whether credit contractions lead to a surge in support for populist parties. To do so, we exploit a sharp and unexpected lending cut by Commerzbank, the second-largest private bank in Germany, at the onset of the global financial crisis of 2007–2008. The shock was driven by losses on the bank's international trading books, and was therefore unrelated to local economic conditions, providing exogenous variation in credit supply. The contraction propagated unevenly across Germany, affecting more strongly those counties with greater pre-existing exposure to Commerzbank. Using data on the bank-firm relationships, we estimate the exposure of each German county to the bank, and therefore to the credit shock.[3] We then combine these data with individual-level survey data to study the effect of this shock on the political preferences of German voters. This approach allows us to compare changes in the preferences of voters exposed to the shock with different intensity.

We find that an increase of one standard deviation in exposure to the credit shock raises individuals' support for populist parties by 0.5 percentage points. This estimate corresponds to an increase of around 16 percent relative to the baseline level of populist

[1] A recent study observed that a quarter of the world's nations, responsible for roughly 95% of global GDP, are led by populist governments (Funke, Schularick and Trebesch, 2023).

[2] For example: 'Populism is the true legacy of the global financial crisis', *Financial Times*, 30 August 2018. 'From Trump to trade, the financial crisis still resonates 10 years later' *New York Times*, 10 September 2018.

[3] We discuss in detail the coverage of firm-level data, drawn from Amadeus, in Section 4.1.

preferences observed in our survey data. This result supports the view that the credit crisis contributed to the electoral rise of populist parties. A simple back-of-the-envelope calculation helps gauge the aggregate magnitude of the effect. Assuming a linear relationship between exposure and political support, and extending the average treatment intensity observed in our sample to the entire country, the credit shock would account for about 8 percent of the overall increase in the vote share of populist parties in German federal elections between 2009 and 2017.

The findings in this paper provide a novel perspective of the rise of populism in advanced economies. Our evidence indicates that the effect of credit shocks on populism does not necessarily stem only from voters' irrational attraction to an anti-establishment rhetoric. Rather, we show that voters are more likely to reward parties combining populist rhetoric with a strong focus on the banking crisis rather than parties relying on populist rhetoric alone.[4] We achieve this result by separately analysing parties' attention to banking issues and their degree of populist rhetoric. We do so by applying text analysis to the parliamentary speeches delivered in the Bundestag, Germany's federal parliament. Using a semi-supervised machine learning approach, we construct yearly party-level measures capturing both the salience of banking-related topics and the intensity of populist rhetoric. This approach allows us to move beyond a binary classification of populism (which we still use for robustness) towards a continuous, data-driven measure.

We further explore the mechanism linking the credit contraction to rising demand for populism. Perhaps counterintuitively, we find that voters did not change their preferences due to the impact of the shock on their personal finances. Individuals were shielded by the support of the government-owned development bank KfW and by the features of the German banking system that facilitate bank-switching for households.[5] The main economic effect of the shock was rather on local economic activity, which translated into

[4]The salience of topics that populists leverage to secure approval among voters is pivotal also in Ahlquist, Copelovitch and Walter (2020) on the bailout of homeowners with mortgages in Swiss francs.

[5]For more details, see Huber (2018).

a persistent decline in local gross domestic product (Huber, 2018). We observe that, in the more exposed areas, the local economy fails to converge back to pre-crisis levels even nine years after the shock. The impact of the shock on populism is therefore explained through a socio-tropic mechanism, whereby voters reacted to the perceived deterioration of the economic conditions in their local community rather than to their personal losses (Colantone and Stanig, 2018). The collective sense of economic malaise shifted political preferences toward parties promising to fight the establishment responsible for the crisis. This finding echoes Cerqua, Ferrante and Letta (2023), that show that voters in areas that struggle to recover from sudden local shocks turn to authoritarian and populist parties, with the mechanism running through institutional distrust. It is also in line more broadly with the theoretical model developed by Panunzi, Pavoni and Tabellini (2024), showing that aggregate economic shocks induce disappointment among voters, making them become 'risk lovers' and therefore more attracted to populist candidates.

Beyond the aggregate mechanism, we ask what determines why some voters react more strongly than others to the same shock. To do so, we identify the individual response to the credit shock by comparing average outcomes of individuals exposed to different degrees of the shock before and after its occurrence. We then use supervised machine learning to predict heterogeneous responses based on a broad set of characteristics. We find that labour market history plays the central role in shaping support for populism. In particular, aggregate unemployment conditions matter more than individuals' current labour market status or education, suggesting that those persistently marginalised in the labour market are especially likely to react against the political status quo.

Our paper contributes to the literature in several aspects. Its main contribution is that it provides evidence of the effect of banking crises on the rise of populism. Existing research has focused on different economic drivers of populism. These include economic insecurity (Guiso et al., 2019, 2024; Garro, 2021), labour market distress (Algan et al., 2017; Hobolt and de Vries, 2016; Barros and Santos Silva, 2019; Dehdari, 2022), and

austerity (Fetzer, 2019; Fetzer, Sen and Souza, 2020; Galofré-Vilà et al., 2021; Dal Bó et al., 2022).[6] Our paper contributes to this literature by studying the persistent economic distress generated by the credit contraction during and after the Great Recession, and how this translated into economic insecurity and political backlash. More broadly, our findings provide causal evidence on how long-lasting disruptions in local credit markets can shape political preferences in advanced democracies.

Our study adds to the relatively smaller literature on the political consequences of disruptions in bank credit specifically. Antoniades and Calomiris (2020) examined the impact of a household mortgage credit contraction in the US on voters' preferences in the presidential election of 2008. However, given the two-party structure of US politics, they necessarily focused on the impact of the contraction on the support for the incumbent versus his opponent, and not on the broader support for populist candidates. Doerr et al. (2022) showed that the 1931 German banking crisis fuelled the support for the Nazi Party, in particular in areas more exposed to *Danatbank*, the bank at the heart of the collapse. This evidence, combined with ours, support the thesis for which banking crises can be key driver of support for anti-establishment political movements. However, this paper focuses on the surge of a specific party, not on the broader surge of populist parties. In our paper, we provide a more detailed account on the impact of credit contractions on different populist parties (including the right-wing party Alternative for Germany, and

[6]Other contributions focus on the political consequences of historical crises (Funke, Schularick and Trebesch, 2016; de Bromhead, Eichengreen and O'Rourke, 2012; Doerr et al., 2022), public finance accountability (Sartre, Daniele and Vertier, 2020), foreign-currency mortgages and household financial distress (Ahlquist, Copelovitch and Walter, 2020; Gyöngyösi and Verner, 2022), and credit cycles or lending shocks (Herrera, Ordoñez and Trebesch, 2020; Braggion, Manconi and Zhu, 2020; Antoniades and Calomiris, 2020). A related strand of the literature focuses on long-run structural transformations that generate economic winners and losers independently of recessions. In particular, globalisation and trade (Autor et al., 2020; Dippel et al., 2022), as well as technological change (Frey, Berger and Chen, 2018; Anelli, Colantone and Stanig, 2019), may increase support for nationalist and far-right parties among those adversely affected in the labour market. Alternative explanations stress the interaction between identity (Mukand and Rodrik, 2018), culture and social capital (Norris and Inglehart, 2019; Autor et al., 2020; Giuliano and Wacziarg, 2020; Ferrara, 2022), economic factors and education (Gethin, Martínez-Toledano and Piketty, 2022), as well as the role of moral values (Enke, 2020), immigration (Alabrese et al., 2019; Dinas et al., 2019; Steinmayr, 2021), and the media (Campante, Durante and Sobbrio, 2018; Cantarella, Fraccaroli and Volpe, 2023). See Zhuravskaya, Petrova and Enikolopov (2020) and Guriev and Papaioannou (2022) for broader reviews of this literature.

the left-wing party Die Linke). Moreover, by introducing a new, continuous measure of populist rhetoric, alongside a more traditional binary party classification. In addition, we measure our main outcome variable using individual-level survey data. This allow us to identify the mechanism at the voter level and to track whether voters react to their own personal financial losses or to the overall deterioration of their local economy, a dimension absent from both studies.

Our contribution is not only substantive but also methodological. Our approach mitigates the gap identified by Guriev and Papaioannou (2022) in connecting the demand and supply sides of populism. We reduce this gap with our continuous text-based measure of populism that moves beyond binary party classifications [7] and combining it with individual-level political preferences. This allows us to distinguish not only different intensities of populist rhetoric, but also differences in the extent to which populist parties focus on banking-related issues.

The remainder of this paper is organised as follows. The next section describes the origin of the credit shock, the political spectrum and the origin of populism in Germany. Section 3 discusses our identification strategy. In Section 4, we outline the data we use to estimate our model. Sections 5 and 6 describe the methods we adopt to measure the credit shock and populism. Section 7 presents the main results and a number of robustness checks. Sections 8 and 9 discuss a mechanism and the decomposition of the main contribution. The final section concludes.

[7] Related text-based approaches include dictionary methods in Enke, Rodríguez-Padilla and Zimmermann (2023), Hassan et al. (2019), Cantarella, Fraccaroli and Volpe (2023), and Breyer (2022), as well as word-embedding approaches in Gennaro, Lecce and Morelli (2024).

2 Background Information

2.1 The Origin of the Credit Shock

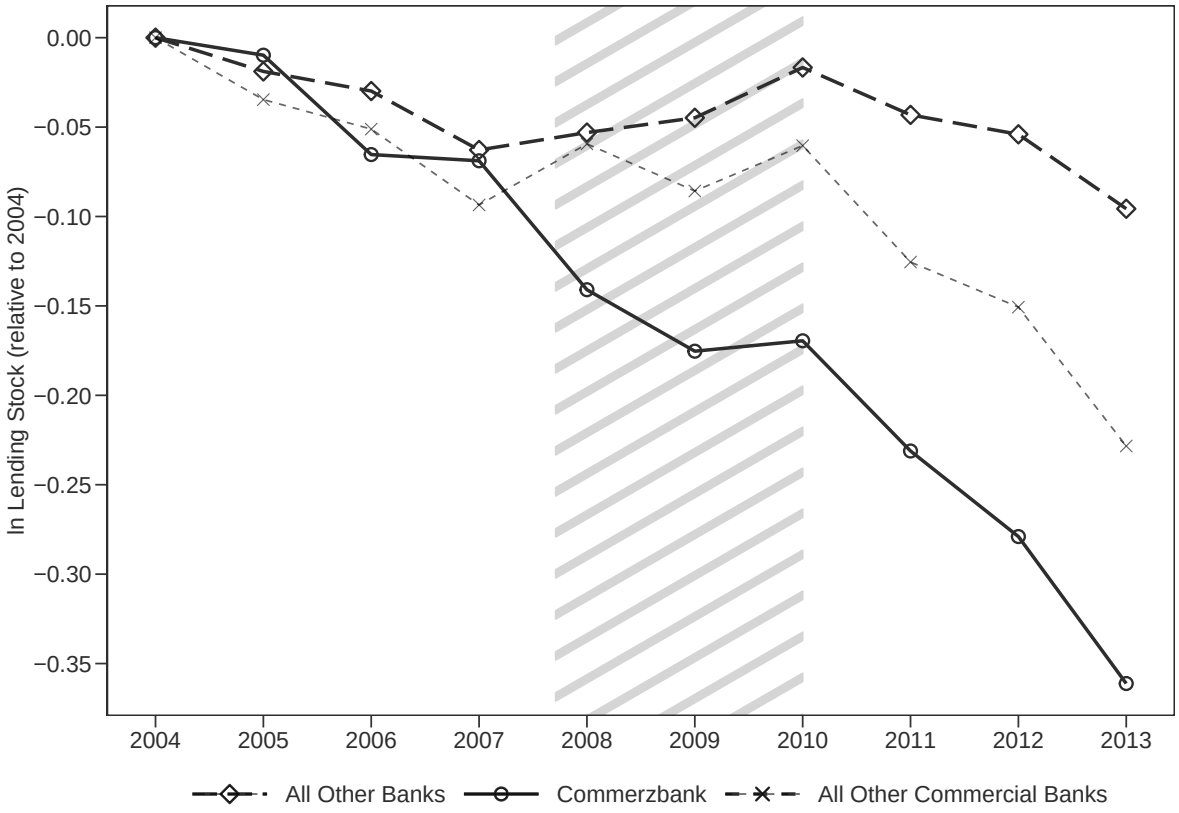
Our aim is to investigate the effect of negative credit shocks on support for populism. The main challenge is to overcome the potential omitted variable bias that could affect this relationship. Omitted variables may simultaneously affect changes in credit and populism, leading to a spurious correlation between lending and populism, even if the true causal effect of the credit shock was null. To this end, we need to identify a shock that was generated exogenously and had potential repercussions on the preferences of German voters. We focus on the imported lending cut suffered by Commerzbank, the second-largest private German bank, in 2008-2009. The lending cut is particularly fit for our research purpose, as it stemmed from losses on the bank's international trading books, and was therefore driven by exogenous causes. This peculiarity allows us to compare German households that were more exposed to the Commerzbank lending cut with those that were less exposed.

Commerzbank is the second-largest private bank in Germany by the total value of its balance sheet, and at the time of the shock it was in charge of around nine per cent of the total bank lending to German non-financial customers.[8] As a universal bank, income earnings come from investing in international markets, interest income from lending and non-interest income from trading.

In Figure 1 we show the natural logarithm of the lending stock of German Banks to non-financial customers. The figure shows that, in 2008 and 2009, lending by Commerzbank declined sharply relative to all other banks, whereas it shows a parallel trend in the period preceding (*i.e.* until 2007) and following (from 2011 onwards) the shock. This difference from the rest of the German banking sector is related to the significant exposure of Commerzbank's trading portfolio to international finance, especially related to invest-

[8] Taken from the rankings in '*Top 100 der deutschen Kreditwirtschaft*' of die.bank, in absolute terms at the time of the shock and among private banks at the current state.

Figure 1: The Lending Stock of German Banks: Commerzbank vs. All



Notes: The graph shows the ln lending stock to German non-financial customers, relative to the year 2004, in 2010 billion euros. Data for Commerzbank include lending by branches of Commerzbank and Dresdner bank, summing their lending stock for the years before the 2009 Dresdner Bank takeover, using information from the annual reports. For all other banks, data come from Deutsche Bundesbank on German banks and subtract lending by Commerzbank. For all other commercial banks, lending stock of Commerzbank, the savings banks, the Landesbanken, and the cooperative banks is removed. The striped area highlights the period during which the Commerzbank lending cut takes place, anticipating the general downturn in the domestic credit. The area starts before 2008 as the lending stocks are at closing date. Replicated from data and calculation in Huber (2018).

ments in asset-backed securities linked to the United States subprime mortgage market, as well as the bank's exposure to the insolvency of Lehman Brothers and the bailout of the Icelandic banks. Given this exposure to foreign securities markets, Commerzbank incurred significant losses on its trading portfolio – decreasing the equity capital by 68 per cent during this period – and reacted by cutting its loan supply to the internal economy, to fulfil Basel II regulations and to reduce risk exposure to be able to access funding markets again.[9] Therefore, the lending cut was completely unexpected and unrelated to changes in the local economy.

2.2 Germany's Party System

Germany is a federal parliamentary republic where legislative power is vested in the *Bundestag* (the German parliament) and the *Bundesrat* (the representative body of the Länder, Germany's regional states). After the reunification, Germany has been a multi-party system with Christian-liberal dominance characterised by ruling coalitions, which is particularly interesting for us because it expresses the heterogeneity of the political system and of populism in Germany compared to the US two-party system. At the time of the credit shock, a grand coalition between the centre-right alliance of Christian Democratic Union (CDU) and Christian Social Union of Bavaria (CSU) and the centre-left Social Democratic Party (SPD) was in charge and co-governed the bailout of Commerzbank. Based on the classification we describe in detail in Section 6.1, we identify three different populist parties on the German political spectrum: (i) Alternative for Germany, (ii) The Left, and (iii) the National Democratic Party of Germany.

First, Alternative for Germany (in German, *Alternative für Deutschland*, AfD) is a far-

[9]The German government fund Soffin supported Commerzbank twice, on 3rd November 2008 and on 8th January 2009, but it was unable to entirely prevent the lending cut. Commerzbank managed to fully recover swiftly by mid-2011 by repaying around 80% of the Soffin equity. The shock was not triggered by the acquisition of Dresdner Bank at the beginning of 2009, as it had been planned way ahead, since 2006. Huber (2018) provides a comprehensive overview of the exogenous lending cut implemented by Commerzbank in the selected years.

right party that won 12.6 per cent of the vote in the 2017 German election and became the first new party to enter the Bundestag since the 1990s. The history of the party can be divided into two phases. During the first phase, the party was a conservative right-wing party embracing economic liberalism. The second phase started in 2015 and saw the transformation of the AfD into a far-right, anti-immigration party. It was in this second phase that the AfD reached its electoral peak in the 2017 election. When it was founded, AfD was not a traditional far-right party. It was a soft Eurosceptic party with an economic ideology that combined economic liberalism with socially conservative policies (Arzheimer, 2015). At the time, the party leadership encompassed conservative members of the German elite, including economists, lawyers and former centre-right politicians, but it also attracted far-right supporters from the very beginning. Already in this first phase, the AfD's stances were more anti-migration and market liberal than those of the centre-right CDU (Jankowski, Schneider and Tepe, 2017; Ceyhan, 2016). According to Schmitt-Beck (2017), in 2013 the AfD attracted two types of voters: those who opposed the euro bailout packages and those who opposed immigration. The AfD's ideological U-turn took place in 2015, when its leader and many supporters left the party. In that year, the party shifted its focus from the euro crisis to migration, and it took strong anti-migrant and anti-Muslim stances. These topics became prominent in the party's political communication as well (Müller and Schwarz, 2021; Doerr, 2021). Based on a text analysis of the party's Facebook posts between March 11th, 2013 (the day they started using Facebook) and September 24th, 2017 (the day of the federal election), Arzheimer and Berning (2019) show that the euro and Greece were among the most prominent topics in the AfD's online communication until 2015. In the second half of 2015, asylum and immigration policies gained traction and have remained prominent since then. This change of focus reflected the AfD's transformation from a single-issue party to a populist radical-right party (Arzheimer and Berning, 2019).

Second, The Left (in German, *Die Linke*) is a traditional radical left party. It emerged

in 2007 from the merger of two leftist parties: the Party of Democratic Socialism (the successor to the German communist party) and the WASG (which broke away from the left wing of the Social Democrats). The main electoral basis of the Party of Democratic Socialism was in East Germany, whereas the WASG drew its support from the West. The parties decided to unite into a single party, The Left, following the disappointing result in the 2005 North-Rhine Westphalia election, when neither of the two managed to poll more than the five percent electoral threshold. Coffé and Plassa (2010) used quantitative text analysis to compare the Left's manifesto with that of its two predecessors on economic policies. They found the agenda of the Left to be closer to that of the Party of Democratic Socialism on economic issues. The Left recorded its strongest performance in the West during the 2009 federal elections, held just one year after the Commerzbank lending cut, capitalising on its anti-establishment and anti-bank rhetoric.

Lastly, the National Democratic Party of Germany was founded in 1964 as the natural successor to the German Reich Party, a nationalist far-right organisation operating in West Germany. It has been regarded as basically a neo-Nazi organisation.

3 Identification

We use a difference-in-differences identification strategy, with heterogeneity across groups and continuous treatment, using two-way fixed effects for inference. In particular, we compare long-term outcomes across counties with different exposure to the credit shock by estimating the following reduced form relationship:

$$y_{ikt} = \delta_k + \lambda_t + \beta \cdot (Exposure_k \times Post_t) + \boldsymbol{\gamma} \boldsymbol{X}_{ikt} + \varepsilon_{ikt} \tag{1}$$

where y_{ikt} denotes the outcome of interest for individual i resident in county (*kreise*) k at time t. Depending on the model, y_{ikt} captures individual preferences towards populist parties or preferences in respect of a specific party. $\boldsymbol{X}_{ikt}$ are observed unit-specific controls

at individual and household level (and specific regional controls in stricter specifications). δ_k and λ_t are county and year fixed effects, respectively.

The main variable of interest, $Exposure_k$, is the heterogeneous pre-shock exposure of a county to the credit shock. Such exposure is measured as Commerzbank dependence, which is computed based on the share of firms in a county that have a bank relationship with Commerzbank. We describe this variable in more detail in Equation (4) in Section 5. $Post_t$ is an indicator variable that equals one for all the years following 2008, and zero for all other periods. To account for the fact that we measure our variable of interest at county level, whereas the outcomes are at individual level, we cluster standard errors at county level (Bertrand, Duflo and Mullainathan, 2004). Throughout the analysis, we use individual-year-specific design weights in our repeated cross-sectional data to overcome survey stratification, sampling and non-response rate.[10]

The coefficient of interest, β, measures the average treatment effect on individuals in counties with higher pre-shock exposure to Commerzbank relative to those in counties with lower exposure beforehand. In a setting with a continuous treatment, however, β does not correspond to a single causal parameter, but combines two distinct effects (Callaway, Goodman-Bacon and Sant'Anna, 2025): (i) the *level* effect, which equals the difference between a unit's potential outcome under exposure and its untreated potential outcome, and (ii) the *slope* effect, which corresponds to the causal response to an incremental change in the exposure (Angrist and Imbens, 1995). Under standard assumptions (random sampling, continuous support of the treatment, and no anticipation),[11] β is equal to a positively weighted average of the average causal response (ACR) parameters across

[10] In a robustness specification presented in Table **??** in the Appendix, we apply longitudinal weights to a balanced panel that substitutes for our main rolling panel sample, where the weights are calculated by multiplying initial sampling weights by the inverse staying probability at the individual level throughout the period of the panel.

[11] These assumptions are achieved by design. The original assumption supporting continuous treatment calls for the presence of never-treated units, such that the probability of being untreated is higher than zero. However, the decomposition of β that establishes the indicated treatment effect continues to apply by extending it to a null probability of being untreated, or no stayers. This is our case under the preferred proxy measure of the exposure to the credit shock.

different exposure levels relative to a baseline level effect given by the weighted average treatment effect on the lowest exposure. The weights depend on the variance-weighted average of all the comparisons among units that experience a different exposure.[12]

Identification requires a parallel trends assumption: in the absence of the Commerzbank credit shock, individuals living in counties with different levels of pre-shock exposure would have experienced similar trends in populist preferences. In other words, any post-shock differences across counties with higher versus lower exposure can be attributed to the credit contraction rather than to pre-existing differences in trends. If this condition does not hold, the estimated effect would reflect both the impact of the shock and underlying differences across counties, leading to biased estimates.

While the parallel trend assumption cannot be directly tested, in Section 7, we provide supporting evidence using dynamic specifications. In particular, we estimate an event-study model that allows treatment effects to vary flexibly over time and assess the presence of differential pre-trends across counties with different exposure levels. For example, if counties with high exposure already exhibited increasing support for populist parties prior to the shock, the event-study coefficients would show a positive trend in the pre-shock period. By contrast, flat pre-shock coefficients provide support for the parallel trends assumption. To this end, we discretise the continuous exposure measure using cut-offs and estimate the following specification:

$$y_{ikt} = \delta_k + \lambda_t + \sum_{r \neq 2008} \beta_r \cdot [\mathbf{1}\left(Exposure_k > s\right) \times \mathbf{1}\left(t = r\right)] + \boldsymbol{\gamma} \boldsymbol{X}_{ikt} + \varepsilon_{ikt}. \tag{2}$$

The coefficients β_r trace out the evolution of the treatment effect relative to the omitted

[12] Callaway, Goodman-Bacon and Sant'Anna (2025) add that β might not equal the natural overall average causal response since the weights of each marginal effect are generally not equal to the density of the treatment conditional on positive treatment. However, when the dose distribution is symmetric and closer to the normal, these weights are closer or identical to weighting the average causal responses to the population distribution of the treatment, leading to the true effect. Figure 2 shows a slightly positive skewness in the distribution of the treatment, which implies that the weights might be closer to the mean dose than their population weights. This is an additional rationale for testing the effect on binary treatments assigned at different cut-offs in Table 2, where we see a consistent effect while considering only the interquartile range of the treatment distribution.

year (2008). In the absence of differential pre-trends, these coefficients should be close to zero in the pre-shock period.

In addition, to reduce noise and improve statistical precision, we estimate a more parsimonious specification that aggregates time into three periods: pre-shock (2000–2007), the shock year (2008), and post-shock (2009–2017):

$$y_{ikt} = \delta_k + \lambda_t + \sum_{\tau=1}^{3} \beta_\tau \cdot [\mathbf{1}\left(Exposure_k > s\right) \times \mathbf{1}\left(t \in \tau\right)] + \boldsymbol{\gamma} \boldsymbol{X}_{ikt} + \varepsilon_{ikt}, \quad \tau \in \{[2000, 2007], 2008, [2009, 2017]\} \tag{3}$$

This specification enables us to remove part of the noise when estimating the year-by-year treatment interactions due to sample constraints, since we only have three time periods.

4 Data

We combine multiple databases in order to estimate the effect of the credit shock on populism. This section outlines the features of the data we use and provides some descriptive statistics. We use firm-level data to compute the exposure of each German county to Commerzbank's lending as a proxy for exposure to the credit shock by looking at the bank relationships of each firm. We then cluster the firm-level information at the level of the county based on each firm's location. We then use individual-level survey data to capture political preferences and to control for a rich set of characteristics. We match this information with our indicator of exposure to the shock based on the county where firms and survey respondents are located. This allows us to identify changes in political preferences depending on the degree of exposure to the credit shock. Since our measure of exposure is at county level, we include a number of economic indicators at county level as controls, which we reapply when looking at the economic performance of those counties more affected by the shock. Finally, we use expert surveys and textual data to identify a

party as populist. Given the complexity of the exposure measure and the indicators we construct, we analyse these two last data sources separately in the following sections.

4.1 Firm-Level Data and Bank Relationships

We collect firm-level data for German companies from the database AMADEUS. The database is one of the products in Bureau van Dijk's OSIRIS and provides comprehensive information for approximately 19 million companies in both Western and Eastern Europe. Banks and financial companies are generally excluded from the database, such that the remainder are only commercial companies. Companies are defined in different categories by their operating revenue, total assets and number of employees, which are handled in four separate tables in the database (small, medium, large and very large). The majority of firms are in the first two categories.[13] In addition to information about the firms, the database AMADEUS BANKERS provides cross-sectional information about the bankers the company has relationships with. When this information is available, we are able to match each firm with its bank relationships through a unique identifier.

The company profile data in AMADEUS provide us with useful information enabling us to match them with the county of their location. For the matching procedure, we first use the postcode information (that is available for most of the firms) to allocate firms to the county where they are situated.[14] For the remaining unmatched firms, we use information on their region or city to identify the county where they are located.[15] To harmonise county-level data with individual level data, we consider 2017 administrative

[13] Small and medium-sized firms constitute around 85% of our companies sample. Throughout the analysis, we use a firm sample from July 2020 without firm size category available due to data loss. We retrieved a different vintage of the firm data (March 2022) with firm size indicators, and the two samples are comparable.

[14] We implement the ZIP Code – Official Municipality Key (AGS) matching table using the list provided by suche-postleitzahl.org.

[15] Stepwise, we use strings for the region or the city where the firms are located to run the Jaro-Winkler distance algorithm (Winkler, 1990) implemented in the `stringdist` R package (van der Loo, 2014) to fuzzy match the string contained in the company profile data to the names included in the matching table we have previously created. We manually adjust the remainder of firms that are not perfectly matched and exclude a few firms with unreasonable locations.

district nomenclature, since some counties were merged into existing districts during our considered time window as a result of state reforms.[16] Figure **??** in the Appendix shows the distribution of the number of bank relationships for each firm in the final sample.[17]

The AMADEUS database is a cross-sectional snapshot of firm-level information, of which we use the version downloaded in 2020. A potential concern is that a cross-sectional snapshot may not accurately reflect bank-firm relationships as they existed prior to the 2008 shock. However, this concern does not apply here for two reasons. First, bank-firm relationships are well documented to be highly persistent over time. In a systematic comparison of firm-level bank data sources, Kalemli-Ozcan et al. (2015) document that bank relationships recorded in cross-sectional snapshots are strongly correlated with their historical counterparts precisely because of this stickiness. This is particularly true in Germany, where the Hausbank model implies that firms maintain stable, long-term credit relationships with a primary lender. Second, the validity of our exposure measure is directly supported by its strong correlation with the pre-crisis Commerzbank lending patterns documented in Huber (2018), who uses the same identification strategy and finds quantitatively comparable economic effects.

4.2 County-Level Macro Data

We record several macroeconomic variables at the regional level from Statistisches Bundesamt (DeStatis), Statistische Ämter des Bundes und der Länder (RegionalStatistik) and Bundesamt für Kartographie und Geodäsie (BKG). We retrieve population, size, percentage of foreign citizens, regional GDP, employment, average household income, an indicator variable of whether the county is a rural or an urban area, whether it is a county of

[16] We retrieve the correct county identifiers for the 401 German counties at 31st December 2017 from the SOEPRemote server at DIW Berlin and the Bundesamt für Kartographie und Geodäsie (BKG).

[17] In the Appendix, we also present in Figure **??** the distribution of the number of firm-bank relationships including the financial and public sector firms that we exclude in the calculation of our main proxy of the credit shock in Section 5, the distribution of the number of firms per county (Figure **??** and **??**, excluding or including finance and public sector firms), and their spatial distribution across the country's area (Figure **??** and **??**).

the former German Democratic Republic (GDR) or whether it is exposed to a similar simultaneous crisis as the lending cut performed by Commerzbank (see Puri, Rocholl and Steffen, 2011). The indicator variables are absorbed by the county-level fixed effects in Equation (1).

4.3 Individual Political Preferences

We exploit individual data from the German Socio-Economic Panel (SOEP, Goebel et al., 2019, v36) – a nationally representative longitudinal household survey that, every year, interviews around 30,000 individuals in different samples. We consider waves from 2000 to 2017. The main advantage of this survey is that it provides detailed information about individual and household characteristics and, for our purpose, it annually records political support and intention to vote, which are our main individual-level outcomes. Political support is registered as an indicator variable that is equal to one for an affirmative answer to the question (translated from German) '*Many people in Germany lean towards one party in the long term, even if they occasionally vote for another party. Do you lean towards a particular party?*'. The question is repeated for each considered wave, and its framing gives us a long-term perspective on political preferences. Our key outcome measures the intention to vote for a populist party conditional on political support. The data provide individual preferences for political parties.

We consider individuals who are at least sixteen years old, which corresponds to the eligibility to vote in administrative elections in several counties and allows respondents of the SOEP to answer political questions.[18] Administrative district keys of residence are available at the individual level in the data: this allows us to match each individual with the pre-shock county-level exposure to the credit shock.

[18]For the analysis, we include all the SOEP sub-samples collected during the time span of interest, from 2000 to 2017, which might affect the outcome variable due to oversampling of immigrants, refugees and low-income families. Even though design weights will take these differences into account, in unreported results we test results excluding these particular sub-samples with only negligible differences in the main results.

Table **??** describes the data in our full sample, with the variables relevant to our analysis. Monetary values are adjusted for inflation at 2016 current prices. We derive the annual household disposable income as household market income (defined as post-government income in Becker and Hauser, 2000) plus public pensions and state monetary transfers minus direct taxes and social security contributions, but including the rental value of owner-occupied homes (Grabka and Goebel, 2018). We include an indicator variable for former residence of the individual in the GDR before reunification to control for political preferences towards extremism (*e.g.* see Avdeenko, 2018; Lichter, Löffler and Siegloch, 2020).

5 Measuring Exposure to the Credit Shock

Having assessed the exogeneity of the shock in Section 3, we now need to find a measure to distinguish between subjects who were hit by the shock (treatment group) and subjects who were not (control group). As the accurate degree of exposure to the credit shock at county level is not directly observable, we use the link between firms and bank relationships described in Section 4.1 to compute a proxy based on a weighted average of the firm-level dependence on Commerzbank. For each set of firms F_k in a county k in 2006, we apply the following equation as proposed in Huber (2018):

$$Exposure_k = \sum_{f \in F_k} \omega_{fk} \underbrace{\frac{\#\text{Commerzbank Relationships}_{fk}}{\#\text{Total Bank Relationships}_{fk}}}_{\text{firm-level Commerzbank dependence}} \in (0,1) \tag{4}$$

where $\#\text{Commerzbank Relationships}_{fk}$ is the number of bank relationships of firm $f \in F_k$ in county k that are with Commerzbank branches, $\#\text{Total Bank Relationships}_{fk}$ is the total number of bank relationships of firm f in county k, and ω_{fk} is a firm-county weighting factor. We identify bank relationships with Commerzbank by using string matching. Following Huber (2018), we primarily set it to $\omega_{fk} = 1/F_k$ equally weighting within a

county.[19] We plot the distribution of the measure of county-level Commerzbank exposure in Figure 2, whereas Figure 3 plots the geographical distribution of the exposure to the shock.[20]

To ensure that the firms in our sample were active in 2006, we select only firms established before 2007. Moreover, we exclude firms without industry sector and firms that are engaged in the finance and public sector other than those already excluded by default in the database (*i.e.* banks and insurance companies).[21] Figure 4 shows the distribution of the firm-specific Commerzbank dependence in the firms sample, which is specifically the ratio within the summation in Equation (4), with details on the frequency of the firms' sample.

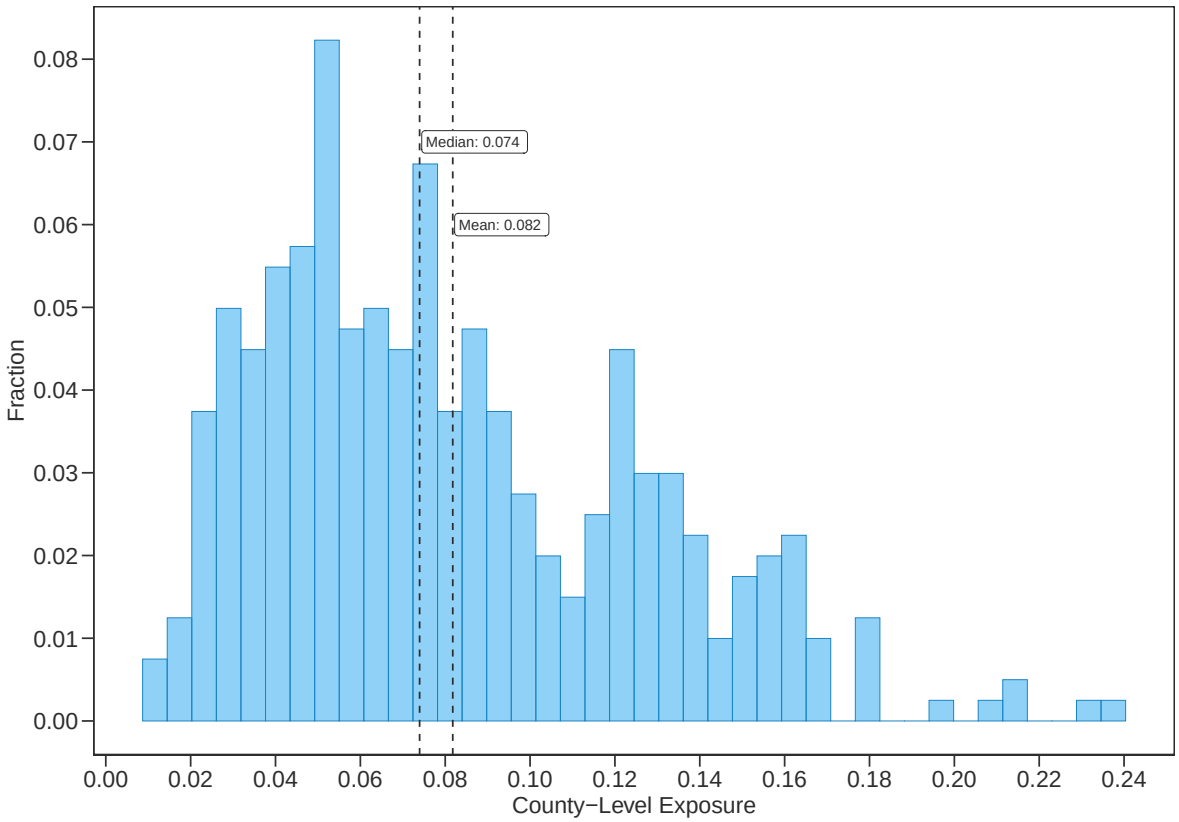
For the measure calculated in Equation (4) to be a good proxy of the county-level exposure to the credit shock, we rely on two assumptions: (i) banks prefer to form relationships with geographically close customers, and (ii) bank relationships are consistently stable over time. While the former establishes a correct relationship between the county-level exposure to the credit shock and the proxy computed in Equation (4), and there is robust support for it in the literature (see Guiso, Sapienza and Zingales, 2004; Degryse and On-

[19] In an unreported extension of the analysis, we weight firms within a county based on the number or cost of employees (payrolls) in 2007, or we match the average number or cost of employees with each firm size category described in Section 4.1. Unfortunately, due to the survivorship bias in AMADEUS, we lose a significant portion of the firm data. Therefore, we decided to run the main analysis using equal weighting, regarding firm heterogeneity to the response to the credit shock as a subject for future study.

[20] From the picture, it is interesting to highlight four major areas of exposure in between which the exposure fades, excluding a major part of Bavaria (except for Munich): (i) East Germany, part of Upper Franconia and Upper Palatinate, (ii) Ruhr Area, (iii) Hessen, and (iv) Hamburg Area. The exposure in East Germany can be explained by the strategy chosen by Commerzbank compared to the other major banks upon Reunification in the 1990s (Klein, 1993). In fact, Commerzbank decided not to take over existing bank branches from the former Kreditbank and mix them with new own branches, but instead exclusively pursued internal growth by creating its own new branches. Its expansion was much slower than for the other two institutions, but it was completed before the 2000s. The other three areas are a by-product of the 'mandated banking zones' imposed by the Allies after World War II to decentralise the former Third Reich financial sector. These areas were chosen for reasons unrelated to economic activity but based on arbitrary decisions by the Allies (Huber, 2018).

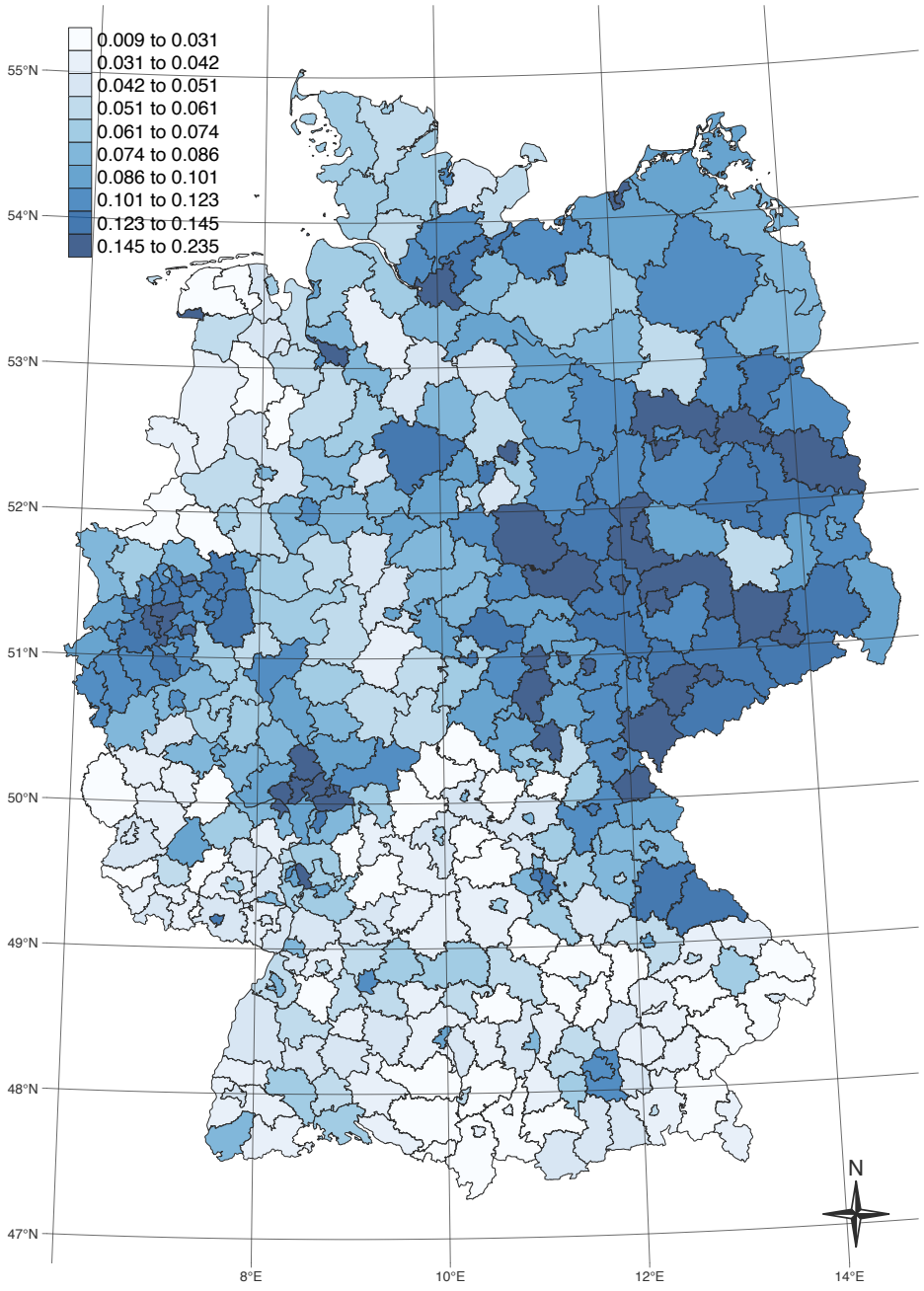
[21] Following Berg, Reisinger and Streitz (2021), we exclude financial services and related industries, including holding companies, industries that in Germany are mostly in the public sector – such as administrative services, education, healthcare, arts, culture and gambling – and activities of organisations and private households. We repeat the exercise in Klapper, Laeven and Rajan (2006) but translated to NACE Rev. 2 codes as in Berg, Reisinger and Streitz (2021). In unreported results, we perform the analysis including those firms, with no significant difference in the main results.

Figure 2: Distribution of County-Level Commerzbank Dependence



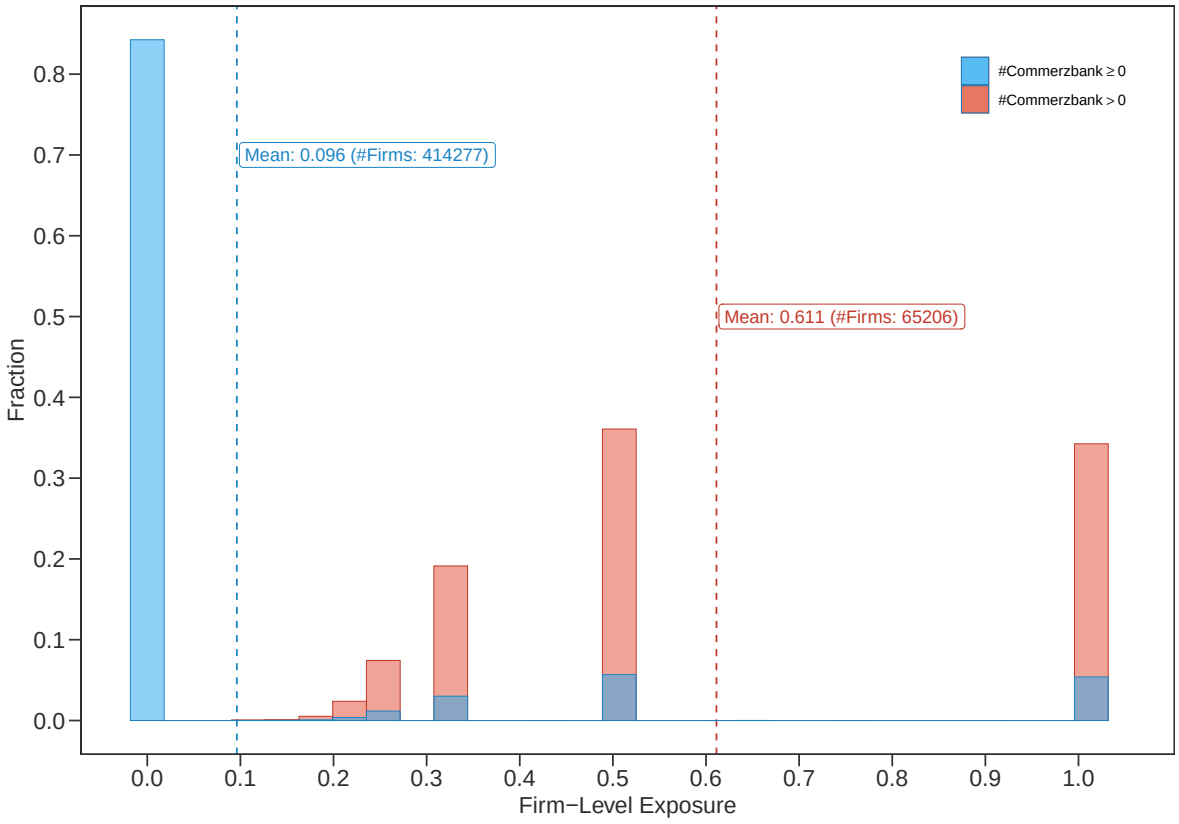
Notes: This figure plots the distribution of the measure of Commerzbank dependence calculated as in Equation (4) at county level in 40 bins. We assign equal weights to firms within each county. We highlight the mean and the median of the distribution with the dashed lines labelled with the exact number. The description of the underlying firms sample is included in Section 4.1 and Figure 4. Source(s): Amadeus, Amadeus Bankers and authors' calculation.

Figure 3: Spatial Variation of the Proxy for the Credit Shock



Notes: This picture maps the distribution of the measure of Commerzbank dependence in Figure 2 calculated as in Equation (4) over the German counties. We assign equal weights to firms within each county. Different values of exposure are binned by deciles. The description of the underlying firms sample is included in Section 4.1 and Figure 4. Source(s): Amadeus, Amadeus Bankers, BKG, and authors' calculation.

Figure 4: Distribution of Firm-Level Commerzbank Dependence



Notes: This picture shows the distribution of the firm-specific Commerzbank dependence in the firms sample. We provide overlapping histograms of a) the unconditional distribution of the firm-level Commerzbank dependence for all firms, and b) the conditional distribution of the firm-level Commerzbank dependence for firms with at least one bank relationship with Commerzbank. The dashed lines indicate the average firm exposure of the unconditional and conditional distribution, respectively. The labels next to the lines indicate the exact mean value and the number of firms involved in the computation. After extracting the firm profiles and matching them with the county code of each firm's location and with the bankers' data as described in Section 4.1, we filter for firms with an incorporation date safely prior to the credit shock in 2007, with a total of 624,258 firms. We exclude firms in the finance and public sector as in Huber (2018) and Berg, Reisinger and Streitz (2021), following Klapper, Laeven and Rajan (2006) but translated for NACE Rev. 2 two-digit codes, for a total of 414,277 firms. Before filtering out firms in the finance and public sector from the sample, we are able to match firms with bank relationships for a total of 946,184 bank relationships, of which 98,990 are with Commerzbank. After filtering out those firms, the total number of bank relationships in the data amounts to 644,265, 65,232 (10.1%) of which are with Commerzbank. Source(s): Amadeus, Amadeus Bankers, and authors' calculation.

gena, 2005), we require the latter to compensate for data limitation, as we only have cross-sectional data for bank relationships based on the last updated information.[22] In support of our assumption, Giannetti and Ongena (2012) and Chodorow-Reich (2014) document the enduring stickiness of bank-borrower relationships. In particular, Chodorow-Reich (2014) shows that, establishing the link between credit market frictions and employment during the financial crisis, pre-crisis clients of banks that restricted lending during the crisis could at no cost switch to borrowing from less constrained banks, but they do not do so due to stickiness in the borrower-lender relationship deriving from asymmetric information. Furthermore, Kalemli-Özcan, Laeven and Moreno (2022) compare different vintages of AMADEUS BANKERS with negligible differences, and Berg, Reisinger and Streitz (2021) use the same data to test Huber (2018) results at firm level.

6 Measuring Populism: Expert Surveys and Text Analysis

Our aim is to identify an indicator that captures support for populist parties, *i.e.* the dependent variable in Equation (1). Defining a party as populist is not easy since populism may rely on different aspects, such as a certain set of policy preferences, ideology, or rhetoric (Norris, 2020; Guriev and Papaioannou, 2022). To account for these nuances, we employ three different, but complementary approaches to obtain comparable indicators of populism at party level. The first method is based on a binary classification of parties as populist based on expert surveys. The second and third are based on semi-supervised and supervised text analysis techniques, respectively.

[22] Unfortunately, bank relationships data for 2006 are not publicly available, nor on request. However, the data provider for the bank relationship data for 2006 exploited in Huber (2018) is the same as in AMADEUS BANKERS.

6.1 Dichotomous Classification using Expert Surveys

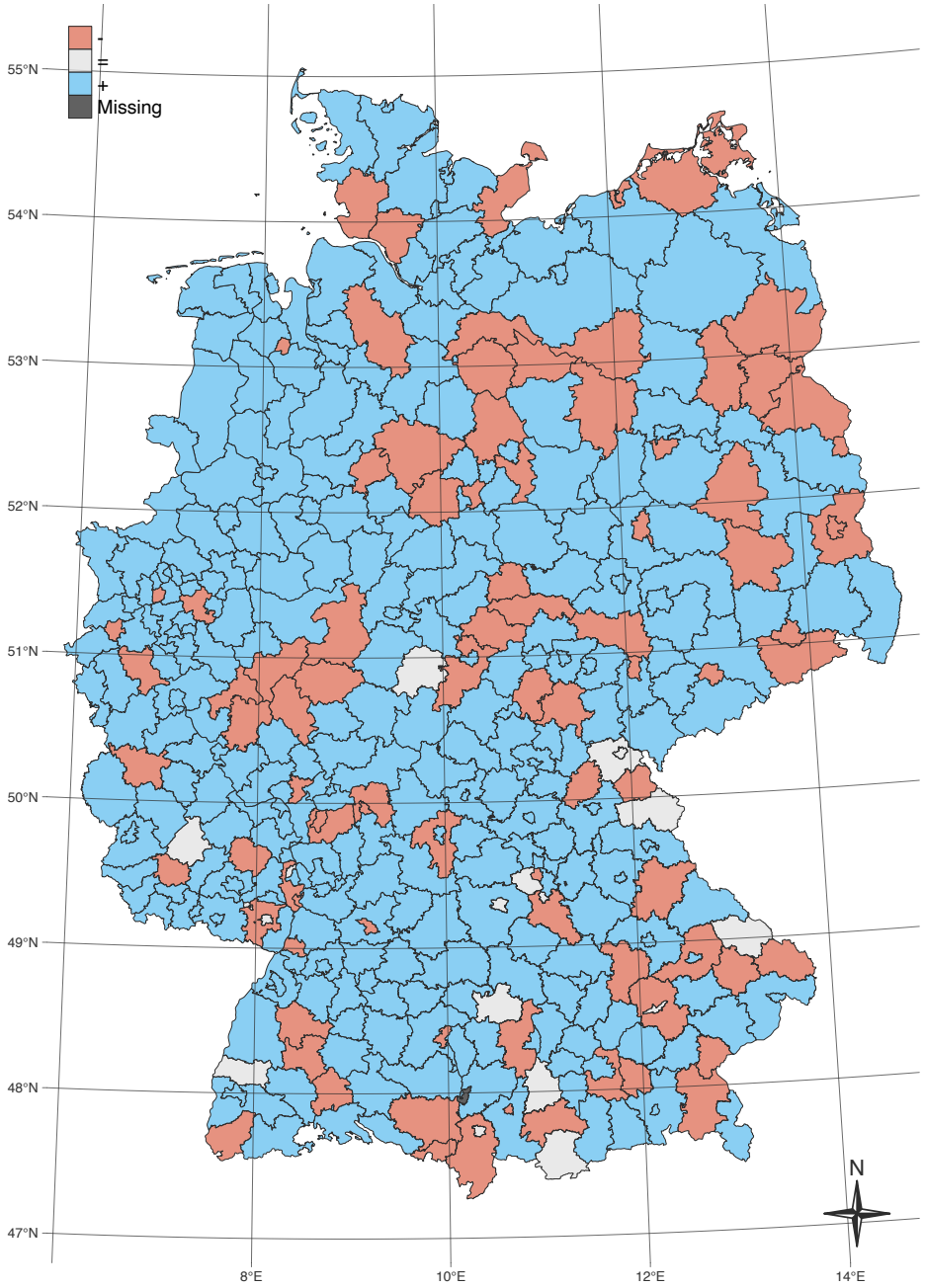
We create an indicator variable that is equal to one when the individual leans towards a populist party. To identify populist parties in Germany, we rely on the *PopuList* proposed by Rooduijn et al. (2019) as in Guiso et al. (2024). The PopuList is a list of populist European parties that won no less than two percent of the vote in at least one national parliamentary election since 1998. The list was peer-reviewed by more than thirty academics. On the basis of these data, we identify as populist the *Die Linke* (The Left) and *Alternative Für Deutschland* (Alternative for Germany, AfD) parties. Since our individual survey data are not constrained by any threshold on the share of votes, we are also able to include on the list of populist parties the *Nationaldemokratische Partei Deutschlands* (National Democratic Party of Germany, NPD), a party that has never won a seat in federal elections, but that features in the classification of Norris and Inglehart (2019) based on the 2014's Chapel Hill Expert Survey (CHES). Figure 5 illustrates the geographical distribution of the difference in the share of populist preferences at county level from the individual party preferences assigned using the expert surveys.

This measure has two main limitations. First, its binary structure only allows comparisons of populist and non-populist parties, but not different degrees of populism. Therefore, this measure is not able to describe whether some parties have more populist stances than others. Second, this indicator is time invariant. As a result, we cannot identify whether the degree of a party's populism changes over time and whether voters' preferences change accordingly.

6.2 Continuous Measurement using Text Analysis

To overcome these limitations, we employ a second measure of populism based on text analysis. We rely on two sources of political textual data. First, we use the text from parliamentary speeches by party representatives in debates in the Bundestag. We use

Figure 5: Difference in Average Populist Preferences after the Credit Shock



Notes: This map shows the German counties with a higher difference in the share of populist preferences after the credit shock based on the relevant question in the German Socio-Economic Panel (SOEP) described in Section 4.3. First, we calculate the sample-weighted mean at county level of the binary question on political support, pooling all the respondents for all years before the credit shock and all years after, and we note the difference in the two shares of populist preferences. Second, we construct an indicator variable equal to one if the difference in the share is above the median of the distribution. We distinguish between counties that present a difference in the share above the median of the distribution and counties below or at the median of the distribution. In Figure **??**, we provide the difference in percentage points before and after 2009 in bins. The sample includes individuals aged at least 16, considers self-reported politically inactive individuals as non-populist, and excludes non-respondents or invalid answers. Non-respondents or invalid answers on political party preferences are only around 2% of the entire sample among those individuals that answer the first question in the affirmative, whereas individuals that refuse to answer the question are around 0.5% in the sample. Source(s): German Socio-Economic Panel (SOEP, Goebel et al., 2019, v36), BKG and authors' calculation.

the *ParlSpeech (v2)* database from Rauh and Schwalbach (2020), which contains the full text corpora of ∼6.3 million parliamentary speeches in nine representative democracies, including Germany. From this source we select the sub-sample of speeches from the German Bundestag in the years from 1991 to 2018. Our sub-sample includes 379,545 speeches, with an average of 13,555 speeches per year from 1991 to 2018. The main limitation of this measure is that, by construction, it necessarily rules out those parties that do not hold seats in the Bundestag.[23]

We compute the degree of populism using seeded Latent Dirichlet Allocation (seeded LDA). Seeded LDA is a semi-supervised, machine learning method used to extract the intensity of a topic from a given set of textual documents (Lu et al., 2011; Watanabe and Zhou, 2020). Seeded LDA works similarly to a classical LDA, which is an unsupervised method used to uncover the latent topics in a text. LDA (Blei, Ng and Jordan, 2003) is a generative probabilistic model based on the assumption that each document contains a mixture of topics and that the words observed in the document in a corpus are generated by latent topics. The main difference between the two approaches is that seeded LDA extracts these topics based on a prior 'seed' of selected terms that capture the object of interest (*i.e.* populist rhetoric in our case). The seeds train the model to extract the latent topics for each document based on the words provided as priors. Watanabe and Zhou (2020) and Ferner et al. (2020) show that this method fixes the inconsistency of topics that is generally produced by LDA. Another advantage of seeded LDA, unlike LDA, is that it does not require the selection of a pre-determined number of topics K. Overall, LDA can be helpful if we want to identify the topics that a text comprises, and we have no priors on them. On the other hand, seeded LDA is preferable in our case since we already know which topics we intend to identify, *i.e.* the topic of banking and the financial crisis and populist rhetoric. We summarise the seeded LDA generative process in the plate diagram in Figure **??** together with the full text analysis workflow we adopt for the computation

[23]The NPD never held a seat in the Bundestag. AfD enters parliament in 2017.

of our measures in Figure **??** in the Appendix.

We first select a seed of words that, together, capture the topics of banking, finance and the financial crisis (we provide the full list of words in Section **??** in the Appendix). This enables us to obtain an indicator of how much each party discusses the topic of the crisis and banking. We intentionally do not focus on keywords related solely to crisis. In this way, we can capture how much each party focused on the topic of banking before the financial crisis and the credit crunch happened. For the sake of brevity, we will refer to this first macro-topic as 'banking and financial crisis' in the rest of the paper. Second, we select a seed of terms that captures populist rhetoric. We take these terms from the populist lexicon composed by Rooduijn and Pauwels (2011) to capture the degree of populist rhetoric of German-speaking parties. This lexicon is particularly interesting as it builds on the definition of populism provided by Mudde (2004, p. 543) as *'a "thin" ideology (Freeden, 1998) that considers society to be ultimately separated into two homogeneous and antagonistic groups, "the pure people" versus "the corrupt elite", and which argues that politics should be an expression of the* volonté générale *(general will) of the people'*. Against this backdrop, populist ideology rests on the belief that the people should govern over the elites due to the people's moral superiority. Guriev and Papaioannou (2022) apply this definition to identify the five most common traits that define modern populist parties: (i) no clear common ideology (in line with the 'thin-centred ideology'), (ii) anti-elite and anti-expert sentiment, (iii) anti-globalisation and anti-EU angle, (iv) anti-pluralism and authoritarian angle, and (v) communication style. Given our focus on political speeches, we are particularly interested in the latter of these traits. Guriev and Papaioannou (2022) further unpack the communication style of populist parties and identify three common patterns in their rhetoric: (i) simplicity of message, (ii) aggressive style, and (iii) social media.[24] The other four elements can help to establish a link between the occurrence of

[24]Many populist platforms owe their success to the use of social media and the use of new communication technology to circumvent the gatekeeping role of mainstream media. Populist parties have been active on social media also in the German context: Müller and Schwarz (2021) show how activities on the AfD's Facebook page fuelled hate crimes against migrants. However, the diffusion of nationwide mainstream

banking crises and the rise of populist parties. Opposition to elites and globalisation can easily fit into a rhetoric that opposes the banking sector and banking more broadly, which is the focus of our lexicon. The list is composed of twenty stemmed terms, such as *elit** and *korrupt**.

We select the top twenty tokens based on the topic-specific posterior probability distribution of the topic model. Figure 6 shows the top twenty terms for the topic of banking and financial crisis (divided into four subcategories: bank, crisis, ECB and finance) and for populist rhetoric based on the populist dictionary.

Based on the top twenty terms associated with each topic through seeded LDA, we compute an indicator that, for each party, captures (i) its focus on the topic of banking, finance and the crisis, and (ii) its populist rhetoric. Formally, we calculate the year-party measure L_{pt} for each party p in year t as:

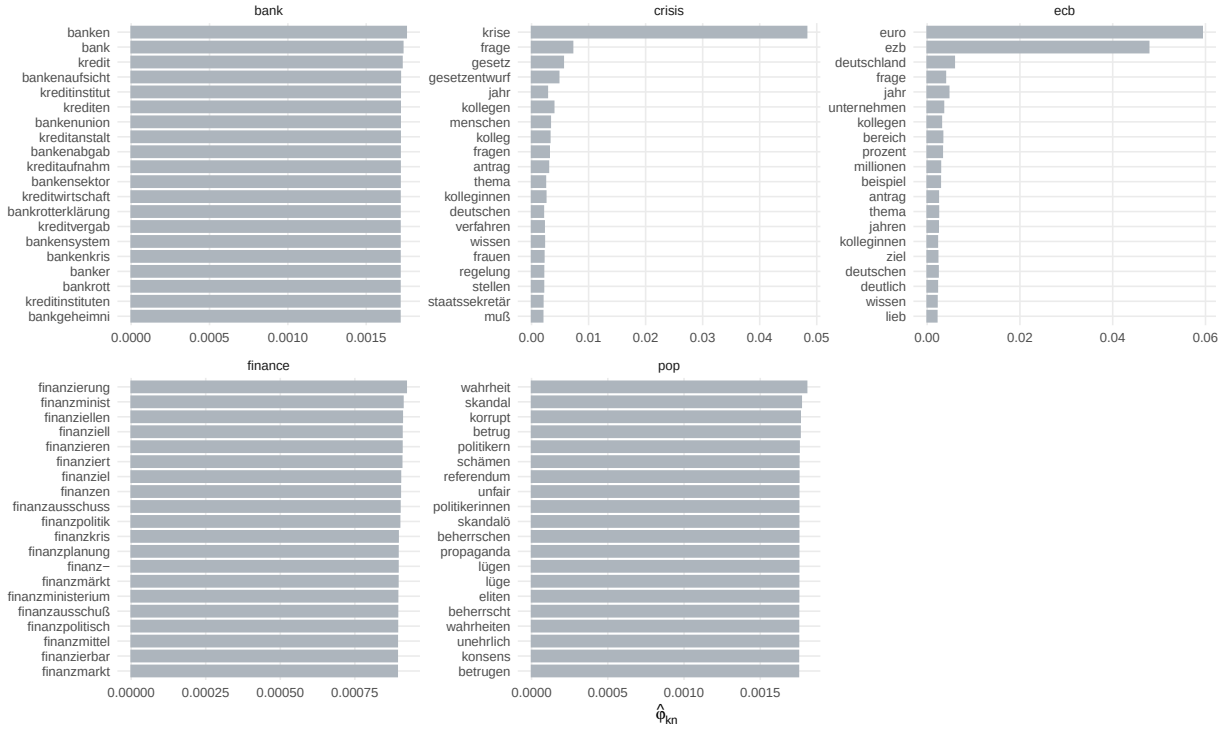
$$L_{pt} = \sum_{d \in D_{pt}} \left[\frac{\sum_{n \in N_d} \mathbf{1}\left(\omega_{dn} \in B_L\right)}{N_d} \right] \; \forall \; L = \{BF, POP\} \tag{5}$$

ω_{dn} is the observed word $n \in N_d$ in document d, while N_d is the per-document d number of words. B_L with $L = \{BF, POP\}$ is a bag of words of $\nu = 20$ tokens with the highest per-topic probability $\hat{\varphi}_k$ for the 'banking and financial crisis' topic and for populist rhetoric defined as the set (**??**) described in section **??** of the Appendix. $D_{pt} \subset \mathcal{C}$ is the collection of speeches for party p in year t of the corpus $\mathcal{C}$. The sum of matched terms, $\sum_{n \in N_d} \mathbf{1}\left(\omega_{dn} \in B_L\right)$ is weighted by the total number of terms in each document, N_d. This allows us to control for variations in the length of speeches.

Figures 7 and 8 show the evolution of the scores by party from 1991 to 2018. Figure 7 shows the focus of each party on the topic of banking and financial crisis over time. We note that all parties increase their attention on banking issues at the beginning of the crisis. The attention most parties devote to the topic peaks in 2010, which marks the

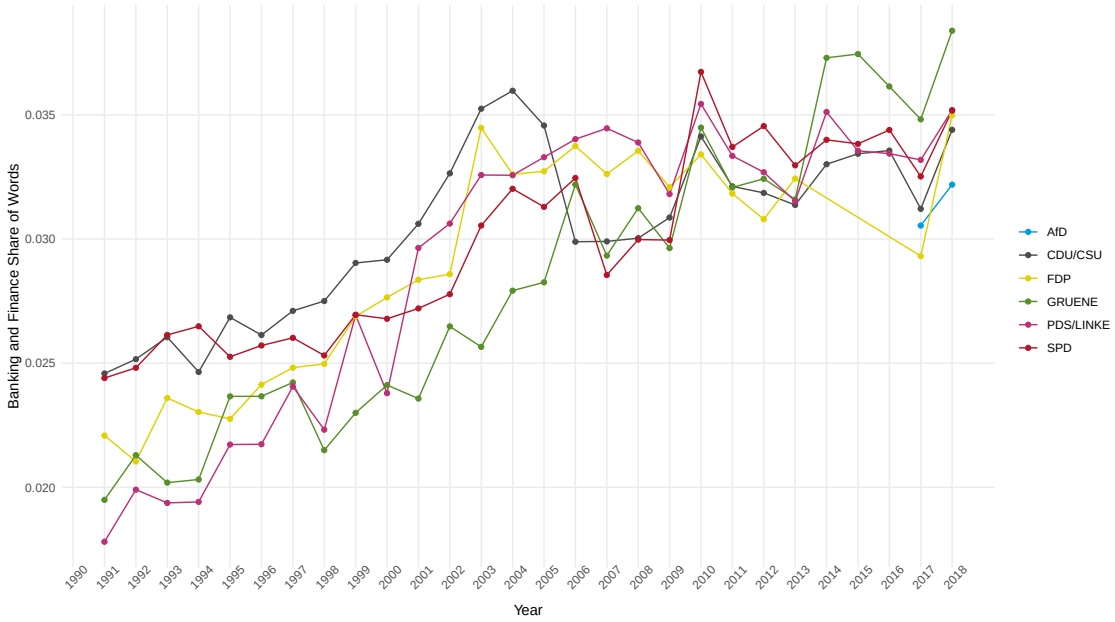
media over local newspapers has also contributed to wider ideological polarisation in Germany from the 1980s up to 2009 (Ellger et al., 2024).

Figure 6: Top Terms by Posterior Probability using seeded LDA



Notes: We use these bar plots to describe the top twenty terms selected using the posterior probability of the seeded LDA models in indicating populist rhetoric and salience of banking and finance-related topics. From left to right, the first four bar plots indicate the top terms for different seeded LDA models on the sub-topics of banking, crises, European Central Bank and finance, from which we select the first five terms to create a score for the salience of banking-related topics. The fifth bar plot shows the top terms for the seeded LDA model on the topic of populism rhetoric, where we select all terms. Seeds for the first four models are uninformative, whereas for the latter we use the Rooduijn and Pauwels (2011) lexicon. The corpus comprises all parliamentary speeches in the German federal parliament from 1991 to 2018 included in the ParlSpeech database (Rauh and Schwalbach, 2020).

Figure 7: Salience of Banking and Finance to Political Discourse by Political Party



Notes: The picture plots the evolution over time of the score for salience of banking and finance in the political discourse for the major political parties in Germany, including those parties that we identify as populist following Rooduijn et al. (2019) in Section 6.1. For each party p, we calculate a measure over time as described by Equation (5) based on a bag of words of 20 tokens, created by each of the top five terms in each subgroup for banking and finance topics as shown in Figure 6. The corpus of parliamentary speeches comprises all parliamentary speeches in the German federal parliament from 1991 to 2018 included in the ParlSpeech database (Rauh and Schwalbach, 2020), the same corpus is used in the topic modelling exercise and the time window goes from 1991 to 2018. FDP held no seats in the German federal parliament from 2014 to 2016, and we therefore have no values for this party in those years. AfD gained seats in the German federal parliament only from the 2017 elections. Over the indicated period, German federal elections took place in 1994, 1998, 2002, 2005, 2009, 2013 and 2017.

beginning of the Eurozone crisis. The low score of the AfD party for banking may appear surprising at first, considering that the party was established by a group of economists with strong stances on the euro crisis and the Greek bailout. However, the party only entered parliament (and hence our sample) in 2017. By that year, AfD had been taken over by its most extremist faction, which focused on topics such as immigration, nationalism and Islamophobia, whereas the economist faction left the party.[25]

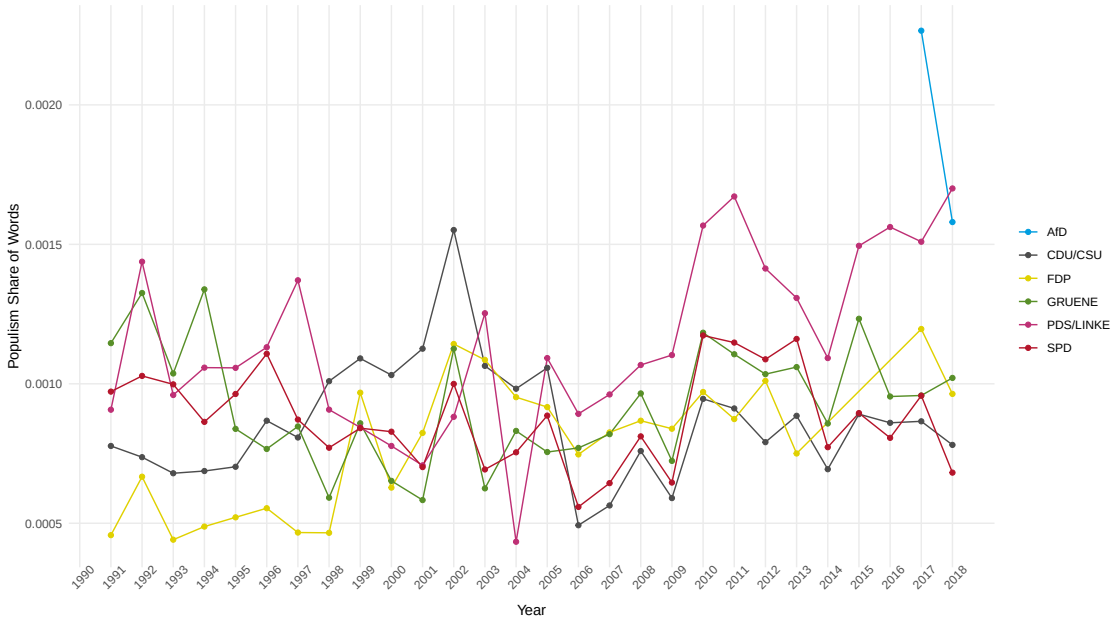
Figure 8 shows the scores for populist rhetoric. We note that the supply of populist rhetoric increases substantially for all parties from 2009 to 2010. However, this change has different intensity depending on the party. The centre-right CDU/CSU and the liberals (FDP) have the lowest scores, followed by the centre-left socialists (SPD) and the Green party (GRÜNE). Populist rhetoric increases sharply for the left-wing party Die Linke, and reaches its peak in 2011 and later on in 2018. From 2006, Die Linke has the highest score in the whole sample until 2017, when the far-right AfD enters the sample. In the last year of our database, Die Linke and AfD are the two parties with the highest degree of populist rhetoric, reflecting the general categorisation of these parties as populist.

Since, for each party, we are interested in comparing (i) its focus on banking and financial crisis, (ii) its populist rhetoric and (iii) the combination of the two, we need a third indicator that provides us with an estimate of the latter. To this end, we create an indicator that we call *Combined* which, for each party and year, equals the average of a party's score for the topic of banking and financial crisis and the party's score for populist rhetoric for each year.

For robustness, we compute the focus on the topic of banking and financial crisis and the degree of populist rhetoric using an alternative text analysis, known as the dictionary approach, or *bag-of-words* approach (Baker, Bloom and Davis, 2016; Shapiro, Sudhof and Wilson, 2020; Ash and Hansen, 2023). We apply the same lexicons that we applied as seeds in the previous approach, in which we define S_V, $V = \{BF^s, POP^s\}$. We then

[25]For a comprehensive description of this transition, as well as of the transition of the AfD's populist rhetoric and issue salience in their speeches, see Cantoni, Hagemeister and Westcott (2020).

Figure 8: Populist Rhetoric in Political Discourse by Political Party



Notes: The picture plots the evolution over time of the score for populist rhetoric in the political discourse for the major political parties in Germany, including those parties that we identify as populist following Rooduijn et al. (2019) in Section 6.1. For each party p, we calculate a measure over time as described by Equation (5) based on a bag of words of 20 tokens, created by the top twenty terms of the seeded LDA model on populist rhetoric as shown in Figure 6. The corpus of parliamentary speeches comprises all parliamentary speeches in the German federal parliament from 1991 to 2018 included in the ParlSpeech database (Rauh and Schwalbach, 2020), the same corpus is used in the topic modelling exercise and the time window goes from 1991 to 2018. FDP held no seats in the German federal parliament from 2014 to 2016, we therefore have no values for this party in those years. AfD gained seats in the German federal parliament only from the 2017 elections. Over the indicated period, German federal elections took place in 1994, 1998, 2002, 2005, 2009, 2013 and 2017.

compute the dictionary-based scores as follows:

$$V_{pt} = \sum_{d \in D_{pt}} \left[\frac{\sum_{n \in N_d} \mathbf{1}\left(\omega_{dn} \in S_V\right)}{N_d} \right] \; \forall \; V = \{BF^s, POP^s\} \tag{6}$$

where the numerator computes the frequency of terms in dictionary S_V that occur in document d, and the denominator weights the frequency by the length in terms of words of the document, N_d. In other words, we compute a similar score to the one in Equation 5. The main difference is that, in this case, we use the raw dictionaries rather than the terms with the highest posterior probability drawn using the seeded LDA. Similarly to the seeded LDA, we also in this case compute a third index based on the average of the dictionary-based score for banking and financial crisis and for populism.

7 The Effect of the Credit Shock on Political Preferences

In this section, we analyse the effect of the Commerzbank's lending cut on the likelihood of supporting a populist party. We present the results from estimating Equation (1) for various specifications in Table 1. In Column 1, we report the estimates of the simplest specification with county and year fixed effects and no additional controls. We find a substantial increase in the likelihood of populist preferences in counties that had higher levels of exposure to the credit shock *ex ante*. That is, the coefficient of $Exposure_k \times Post_t$ denotes that an increase in county-level Commerzbank exposure by one standard deviation makes individuals 0.42 percentage points more likely to prefer a populist party. This counts as a roughly 12.5% increase in populist preferences in the sample, a sizeable effect and statistically significant at 1%. In Columns 2 to 4, we introduce stepwise controls at the individual level (*i.e.* gender, second-order age polynomial, having lived in the GDR, marital status, migration background, categories of occupation, years of education) and household level (*i.e.* size, number of children, home ownership, presence of outstanding

loans) and county-specific controls (*e.g.* ln(share of foreigners), ln(population density), etc.). The coefficient of interest remains fairly stable and significant throughout the specification, varying between a 0.52 and 0.55 percentage point increase in the likelihood of populist preferences for one standard deviation increase in pre-shock county-level exposure, up to a 16.3% increase in populist preferences in the sample. In Column 5, we introduce linear county-specific time trends, partialling out the unobserved variables that move simultaneously with the timing of the credit shock at county level, and obtain consistent results. Table **??** presents estimates of the same above-mentioned specifications in Table 1, but including individual fixed effects throughout, since the SOEP works as a rolling panel and individuals might be present in more than one single wave of the survey. Although we do not prefer results with individual fixed effects since we lose all individuals who are only present in a single wave, this allows us to control for individual-specific unobserved factors as robustness, providing significant estimates that are stable with an increase of between 0.511 in the most basic specification and 0.536 percentage points with full controls.

Figure 9 presents all our estimates from Equation (2) and (3) graphically, qualifying the cut-off for the treatment indicator variable at the median of the treatment distribution. We plot dots indicating year-by-year 'treatment above median' interaction estimates (and bars indicating 95% confidence intervals). These estimates show the difference between individuals located in counties where pre-shock exposure to a credit shock is higher than the median of the county-level distribution and individuals resident in counties up-to-the-median exposed in terms of the likelihood of supporting a populist party in a particular year, relative to the difference in 2008 (first lag before the shock). We can see that, prior to 2008, the difference in populist preferences was not significant and without a clear trend. Then, from 2009 there is a sharp jump in populist preferences among individuals in counties more affected by the shock. Each interaction is positive from 2009, and especially significant in 2010 and 2011. Figure 9 also shows the estimated coefficients of the

Table 1: The Effect of the Credit Shock on Populist Preferences: Difference-in-Differences Results

	(1)	(2)	(3)	(4)	(5)
$Exposure_k \times Post_t$	0.418**	0.517***	0.546***	0.549***	0.542***
	(0.173)	(0.187)	(0.187)	(0.190)	(0.145)
Number of Observations	385,248	362,295	362,122	362,122	362,122
Number of Counties	401	401	401	401	401
Outcome Mean (%)	3.347	3.371	3.373	3.373	3.373
$\sigma\left(Exposure_k\right)$ (%)	4.863	4.861	4.861	4.861	4.861
Adjusted R^2	0.049	0.058	0.059	0.059	0.064
County-Level FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Individual FE	No	No	No	No	No
Individual Controls	No	Yes	Yes	Yes	Yes
Household Controls	No	No	Yes	Yes	Yes
Regional Controls	No	No	No	Yes	No
County Time Trends	No	No	No	No	Yes

Notes: This table reports the results of the estimation of our difference-in-differences specification described in Equation (1), where the dependent variable is our indicator variable of intention to vote for a populist party of an individual i resident in county k at time t, constructed as depicted in Section 6.1, and the variable of interest is the (standardised) exposure to the credit shock of county k, as calculated in Equation (4) using equal weights for all firms within each county, interacted with an indicator variable equal to one for all years after the occurrence of the credit shock. The treatment unit is given by the individuals within each cell (k, t). Sampling weights and county and wave fixed effects apply throughout the results. Column 1 shows the results of two-way fixed effects and no additional controls. Columns 2 to 4 introduce step-wise controls at the individual level, household level, and county-specific controls. Finally, in Column 5 we introduce county-specific linear time trends. The table includes the number of observations, the number of counties, the outcome mean and the standard deviation unit of the treatment for all the results. The coefficients of interest are scaled by 100 to be interpreted as the percentage point increase in the outcome mean of one-σ units increase of treatment after the occurrence of the shock. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively. In Table **??** we show estimates using the same specifications but including individual fixed effects.

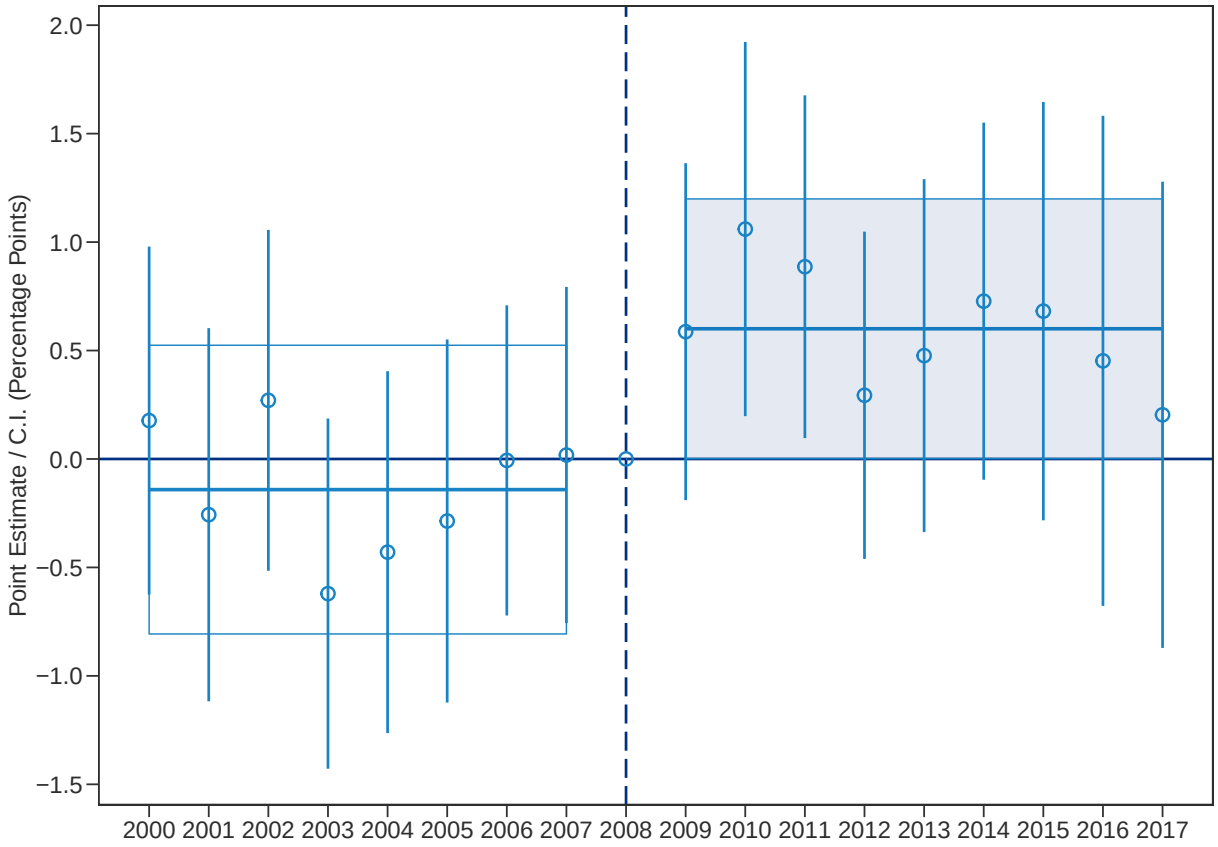
interactions between the more aggregated time period indicator variables and the same treatment indicator variable from estimating the pooled specification in Equation (3). We draw point estimates from the pooled year interactions as horizontal lines with 95% confidence intervals denoted as boxes. Prior to the credit shock, the figure shows close to zero and insignificant differences in populist preferences between individuals resident in more exposed counties and individuals located in up-to-the-median exposed counties beforehand. Conversely, after the occurrence of the credit shock, there is a statistically significant increase in the likelihood of individuals in counties more exposed to the credit shock supporting a populist party.

Figure 10 shows all the estimates in Figure 9, conditioning the results on the covariates in Column 4 of Table 1 in panel 10a and adding individual-level fixed effects on top in panel 10b. We can see from the figures that the common trends assumption holds both with and without conditioning for covariates. Moreover, in Table 2 we estimate Equation (1) substituting continuous treatment with an indicator variable equal to one when the exposure to the credit shock exceeds a certain cut-off of the treatment distribution, each different in every specification in addition to sample exclusion and the inclusion of individual-level fixed effects.

In Tables **??** and **??** in the Appendix, we present results using an indicator variable of political support as the outcome in Equation (1). Even though, in some specifications, we see statistically significant point estimates, we cannot reject the null hypothesis that there are no differences in political support for individuals located in counties with a higher exposure to the credit shock compared to individuals located in counties with a lower exposure after the shock happens. This is particularly justified by the visualisation of the point estimates from Equation (2) and (3), using political support as the outcome. In Figure **??**, we cannot exclude the presence of pre-trends, and estimates of the pooled pre- and post-periods are not significantly different from each other.

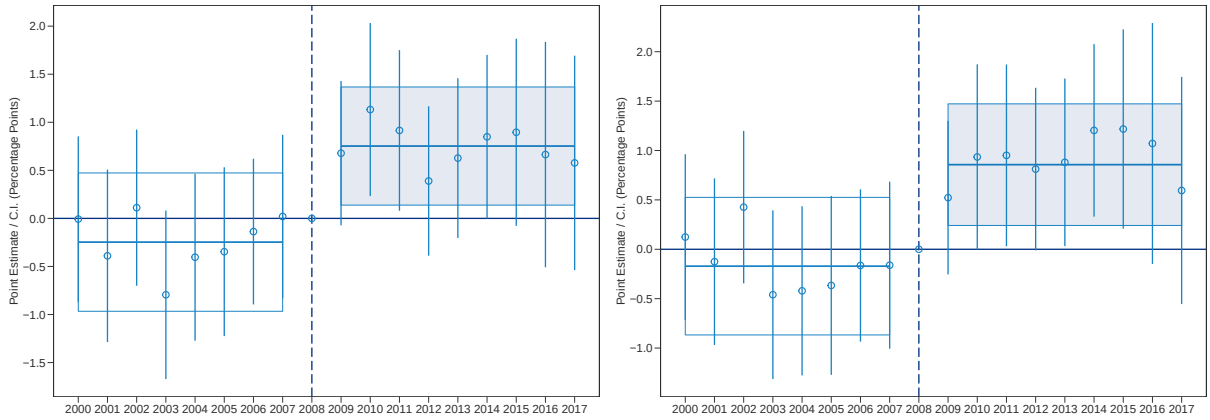
To identify the amplitude of the slope effect on our results, we draw the functional

Figure 9: The Effect of the Credit Shock on Populist Preferences: Difference-in-Differences Estimates



Notes: In this graph, we formally test for differences in the likelihood that individuals would have stronger political preferences towards populist parties after the occurrence of the credit shock, comparing individuals resident in a treated county at time t with individuals resident in an untreated county at time t. A county k is considered as treated after the occurrence of the credit shock when its exposure to the credit shock – calculated as in Equation (4) with firms being assigned equal weights within the county and excluding finance and public sector firms from the calculation – lies above a cut-off s of the treatment distribution. Here, we qualify the cut-off as $s = med\,(Exposure_k)$, the median of the treatment distribution. Regressions are estimated at the individual level on the full sample of 385,248 individual-year observations in 401 counties with no controls, weighted using sampling weights and they include county and year fixed effects. Year regression coefficients of interest from the flexible difference-in-differences design in Equation (2) are interactions between an indicator variable equal to one for treated counties and year fixed effects and are estimated relative to the omitted interaction with the first lag before the occurrence of the credit shock. In the more aggregated differences-in-differences design with three time periods in Equation (3), coefficients of interest are interactions between an indicator variable equal to one for treated counties and between a 2000–2007 dummy ($\beta = -0.141,\ p = 0.677$) and a 2009–2017 dummy ($\beta = 0.600,\ p = 0.049$), estimated relative to the omitted interaction with the first lag before the occurrence of the credit shock. Coefficient estimates of the year interactions are plotted as dots with their 95% confidence intervals indicated by vertical lines. Coefficient estimates of the aggregate interactions are shown by horizontal lines, and their 95% confidence intervals are indicated as boxes, unshaded or shaded for the pre- and post-period, respectively. All the point estimates and 95% confidence bands are re-scaled by 100 to be interpreted as percentage points difference from the baseline, and standard errors are clustered at the county level.

Figure 10: The Effect of the Credit Shock on Populist Preferences: Difference-in-Differences Estimates (with controls)



(a) without individual fixed effects (b) with individual fixed effects

Notes: These graphs are equivalent to the plot in Figure 9 with the sole difference being that regressions are estimated with the addition of the same controls as in Column 4 in Table 1. In Panel 10a regressions are estimated at the individual level for the sample of 362,122 individual-year observations within 401 counties with individual-, household- and county-specific controls. See Table 1 for details on the applied covariates. With this specification, the coefficient of interest to the interactions between the indicator variable for treated counties and a 2000–2007 dummy and a 2009–2017 dummy are $\beta = -0.247$ ($p = 0.501$) and $\beta = 0.753$ ($p = 0.016$), respectively. In Panel 10b, regressions are estimated at the individual level for the sample of 351,304 individual-year observations within 401 counties with individual-, household- and county-specific controls, introducing individual fixed effects, and omitting time-invariant individual-level covariates. With this specification, the coefficient of interest on the interactions between the indicator variable for treated counties and a 2000–2007 dummy and a 2009–2017 dummy are respectively $\beta = -0.171$ ($p = 0.630$) and $\beta = 0.857$ ($p = 0.006$). Both aggregated coefficients are estimated relative to the omitted interaction, with the first lag before the occurrence of the credit shock. All the point estimates and 95% confidence bands are re-scaled by 100 to be interpreted as percentage points difference from the baseline, and standard errors are clustered at the county level. See Equation (2), Equation (3) and Figure 9 for further details on the difference-in-differences design and the regression estimates.

Table 2: The Effect of the Credit Shock on Populist Preferences: Difference-in-Differences Estimates with Binary Treatment

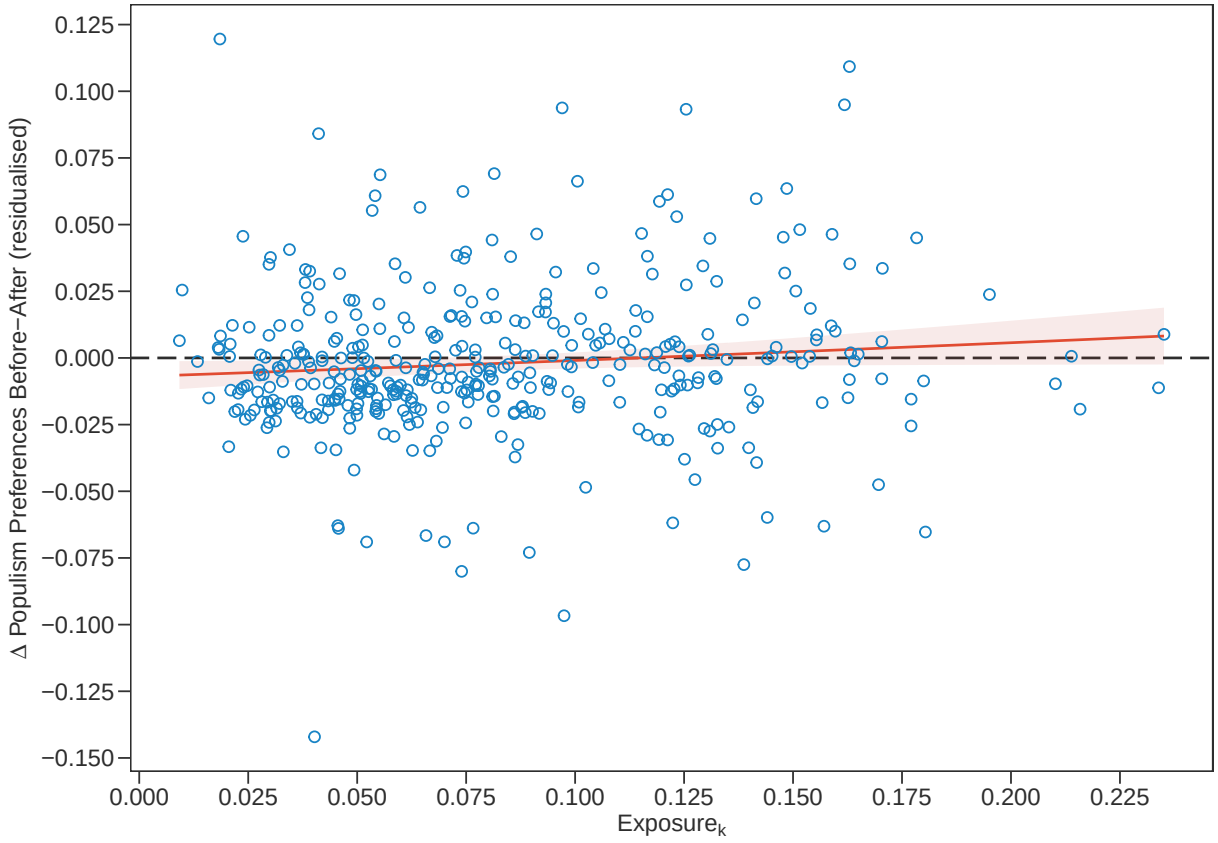
	Median		**75th**		**90th**		**25th – 75th**		**10th – 90th**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\mathbf{1}\left(Exposure_k > s\right) \times Post_t$	0.972***	0.998***	0.889*	1.158**	1.195*	1.175	1.201**	1.410***	1.301	1.290
	(0.352)	(0.329)	(0.497)	(0.490)	(0.701)	(0.755)	(0.551)	(0.523)	(0.821)	(0.782)
Number of Observations	362,122	351,304	362,122	351,304	362,122	351,304	179,100	173,196	71,851	69,441
Number of Counties	401	401	401	401	401	401	208	208	82	82
Outcome Mean (%)	3.373	3.363	3.373	3.363	3.373	3.363	3.669	3.673	3.773	3.73
s (%)	8.949	8.949	13.093	13.093	16.495	16.495	5.834	5.834	3.886	3.886
Adjusted R^2	0.059	0.526	0.059	0.526	0.059	0.526	0.060	0.531	0.069	0.538
County-Level FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Individual Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Regional Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County Time Trends	No	No	No	No	No	No	No	No	No	No

Notes: This table shows the estimates of our difference-in-differences specification described in Equation (1) with variation in the treatment variable of interest. The dependent variable is our indicator variable of populist preferences of an individual i resident in county k at time t constructed as depicted in Section 6.1, while the variable of interest is an indicator variable equal to one if the exposure to the credit shock in county k is above a defined cut-off in the treatment distribution, interacted with an indicator variable equal to one for all years after the occurrence of the credit shock. The treatment unit is given by the individuals within each cell (k, t). Sampling weights and county and wave fixed effects apply throughout the results. Each pair of columns defines the cut-off at which the treatment is assigned, where, in the first column, we use individual- and household-level and county-specific controls, and, in the second column, we absorb for individual fixed effects excluding time-invariant individual controls applied in the specification of the first column. Treatment (control) groups in each specification are those individuals that are located in a county with exposure to the credit shock: (i) above (up to) median of the treatment distribution; (ii) above (up to) the 75th percentile of the treatment distribution; (iii) above (up to) the 90th percentile of the treatment distribution; (iv) above the 75th percentile of the treatment distribution (up to the 25th percentile of the treatment distribution); (v) above the 90th percentile of the treatment distribution (up to the 10th percentile of the treatment distribution). The coefficients of interest are scaled by 100 to be interpreted as the percentage point increase in the outcome mean for individuals located in counties with exposure to the credit shock above the indicated threshold relative to individuals located in counties that have a lower exposure beforehand. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.

form of populist preferences on our continuous treatment. Figure 11 describes the functional form of the outcome variable on the exposure to the credit shock in levels. The dots define the difference in levels of the residualised mean of populist preferences at the individual level before and after the lending cut for each level of the distribution of county-level exposure. It is possible to observe that the difference in populist preferences linearly increases with the intensity of exposure to the credit shock in the linear space, given that the effect on populist preferences follows a linear relationship with the intensity of exposure, as the fitted line shows (the shaded area represents a 95% confidence interval). To more formally test the linearity of the slope effect, in Figure **??** in the Appendix, we highlight estimates of Equation (2) in separate regressions, where we characterise different treatment and control groups based on the position of the exposure to the credit shock in the county where an individual lives within the treatment. Despite the noisy estimates, it is possible to see that point estimates are higher on average for higher treatment.

In addition to testing for the presence of pre-trends in Figure 9, we run a battery of robustness checks based on the specification of Column 4 in Table 1 by restricting the sample for some conditions. Table 3 shows the estimates for each of these robustness checks. In Column 2, we test for conditioning the sample to individuals who are at least 18 years old, finding negligible differences from the baseline estimate in Column 1. In all our results, the binary classification of support for a populist party is unconditional on the indication of a political preference in the underlying question described in section 4.3 to preserve sample size, which is based on the assumption that individuals who state no political support are more likely to prefer a populist party when it comes to non-revealed preferences. We test this assumption in Column 3 by only conditioning preferences for populist parties on individuals with revealed political support, finding robust estimates of a higher magnitude, with an effect of 0.994 percentage points. In addition to this evidence, our results are intention-to-treat estimates and thus provide a conservative lower bound of the true effect since we are estimating a reduced form. Moreover, it is possi-

Figure 11: Functional Form of the Populist Preferences on the Exposure to the Credit Shock: Accounting for Treatment Heterogeneity



Notes: This figure describes the linear relationship between the likelihood that individuals would have stronger political preferences towards populist parties after the occurrence of the credit shock and the distribution of the exposure to the credit shock, calculated as in Equation (4) with firms given equal weight within the county and excluding finance and public sector firms from the calculation. We plot the difference in levels of the (residualised) mean of populist preferences at individual level before and after the lending cut against the distribution of the county-level exposure in levels. Residuals are predicted from a regression of the individual-level outcome variables on individual-, household- and county-specific controls with a sample of 362,122 individual-year observations in 401 counties, using sampling weights, including county and year fixed effects and clustering standard errors at the county level, with the same covariates as in Column 4 in Table 1. After residualising, we compute the mean of the residuals for each level of the continuous treatment before and after the credit shock, and we use the difference in the mean of those values, denoted by the dots in the picture. We filter for three outliers to better depict the linear relationship. The red line illustrates the linear fit, with the shaded area as the 95% confidence intervals. Both variables are kept as levels.

ble that we underestimate the effect of the credit shock on populist preferences due to the tendency of individuals to not reveal extreme political preferences when asked, but rather indicate a preference for the establishment, which is consistent with our findings in Column 3 of Table 3 and the low non-response rate to the political support question indicated in section 4.3. Finally, in Columns 4 and 5, we carry out placebo tests using different timings of the shock, accounting for the large drop in lending stock in 2008Q3-Q4 at the time of Lehman Brothers' default and the anticipation effect of the subprime crisis. We find consistent estimates of smaller magnitude, suggesting a correct selection of the credit shock's timing. As we are exploiting a rolling panel for the analysis, there might be reason for concern about the validity of the results in a more formal sense, where we can actually observe preferences at the individual level, where individuals switch from a more traditional party towards populist parties upon the occurrence of the credit shock. In Table **??**, we address this potential issue. For a restricted sample, we are able to create a balanced panel of individuals that can be followed year by year from 2006 to 2012, 2004 to 2013, and 2000 to 2015, respectively. Of course, this procedure significantly downgrades the statistical power of our analysis, but it allows us to check whether our results still hold when comparing exactly the same individuals with different treatment before and after the occurrence of the shock, applying longitudinal weights. For the first two panels, we do find significant results with a higher magnitude compared to the repeated cross-sectional sample, whereas, for the longest balanced panel, we do not find significant estimates, which is probably related to the dispersion in the number of individuals. Overall, these results strengthen the robustness of our main analysis.

These estimates are based on populism defined as a binary classification of the political parties supported by each individual as being populist or not. While our estimates indicate that being exposed to the credit shock increases the probability of voting for a populist party, it is not clear yet whether individuals do so because of a party's populist rhetoric, because of its focus on banking issues or because of a combination of the two. To

Table 3: The Effect of the Credit Shock on Populist Preferences: Sample Restrictions and Robustness

	Full Sample	Age ≥ 18	Conditional	$\mathbf{1}(t \geq 2007)$	$\mathbf{1}(t \geq 2008)$
$Exposure_k \times Post_t$	0.549***	0.547***	0.994***	0.461**	0.524**
	(0.190)	(0.189)	(0.384)	(0.196)	(0.204)
Number of Observations	362,122	361,619	162,845	362,122	362,122
Number of Counties	401	401	400	401	401
Outcome Mean (%)	3.373	3.375	7.342	3.373	3.373
$\sigma(Exposure_k)$ (%)	4.861	4.861	4.881	4.861	4.861
Adjusted R^2	0.059	0.059	0.174	0.059	0.059
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Individual FE	No	No	No	No	No
Individual Controls	Yes	Yes	Yes	Yes	Yes
Household Controls	Yes	Yes	Yes	Yes	Yes
Regional Controls	Yes	Yes	Yes	Yes	Yes
County Time Trends	No	No	No	No	No

Notes: In this table, we show the results of the estimation using our preferred specification from Column 4 in Table 1 of the difference-in-differences specification described in Equation (1), where the dependent variable is our indicator variable for intention to vote for a populist party of an individual i resident in county k at time t, constructed as depicted in Section 6.1, and the variable of interest is the (standardised) exposure to the credit shock of county k, as calculated in Equation (4) using equal weights for all firms within each county, interacted with an indicator variable equal to one for all years after the occurrence of the credit shock. In Column 2, we restrict the sample to individuals who have reached the age of majority. In Column 3, we vary our outcome variable conditioning populist preferences for answering affirmatively the question described in Section 4.3, relaxing the assumption that people not actively supporting any party are transferring their preferences to more traditional parties when actually voting. In Columns 4 and 5, we perform placebo tests using different timing for the dummy variable indicating the starting period of the shock. The coefficients of interest are scaled by 100 to be interpreted as the percentage point increase in the outcome mean of one-σ units increase of treatment after the occurrence of the shock. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively. A further description of the results is provided in Table 1.

this end, we now replace the dependent variable with the continuous text-based indicator of populism we described in Section 6.2. The first indicator we study is the score based on the seeded LDA estimates and computed on the basis of the parliamentary speeches of the representatives of each party. The estimates of the model for this dependent variable are shown in Table 4. It presents the results for the seeded LDA using the populist lexicon of Rooduijn and Pauwels (2011). As in the previous table, in this case Column 1 also includes county-level and wave fixed effects, as well as basic controls, while in Columns 2 and 3 we progressively add controls at the household and regional level. Columns 1 to 3 show the estimates for the topic of 'Banking and Financial Crisis'. The coefficients capture the probability of an individual hit by the credit shock voting for a party based on the party's focus on the topic of banking and financial crisis. As we saw from the descriptive analyses, many of the terms related to this topic are related to credit and the crisis. The coefficient is positive and significant at the one per cent level, indicating that individuals exposed to the credit crunch were more likely to vote for parties that spoke more frequently about banking and credit issues, regardless of their degree of populist rhetoric.

In Columns 4 to 6, we replace the dependent variable with the text-based indicator of populist rhetoric. The interpretation of these estimates is the same as for the previous columns, but for populist rhetoric. The positive and significant coefficients in Columns 4-6 indicate that individuals exposed to the lending cut are more likely to vote for parties that use populist rhetoric, regardless of their focus on banking issues. This result supports the overall finding presented in Table 1, where the credit shock causes an increase in intentions to vote for populists. The difference is that, here, populism is defined as a continuous – and not a binary – variable, meaning that we are not just comparing populist and non-populist parties, but parties with different degrees of populism. Moreover, here we focus specifically on the use of populist rhetoric in the context of parliamentary debates, whereas binary classifiers are based on a number of factors, such as a party's

Table 4: The Effect of the Credit Shock on Political Preferences: Outcome as Topic Model Scores

	Banking and Financial Crisis			**Populism**			**Combined**		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Exposure_k \times Post_t$	0.060***	0.059***	0.055***	0.137***	0.136***	0.144***	0.070***	0.068***	0.065***
	(0.016)	(0.016)	(0.016)	(0.022)	(0.023)	(0.023)	(0.016)	(0.017)	(0.017)
Number of Observations	161,680	154,562	154,562	161,680	154,562	154,562	161,680	154,562	154,562
Number of Counties	400	400	400	400	400	400	400	400	400
$\bar{y}$	3.181	3.184	3.184	.09	.091	.091	1.636	1.637	1.637
σ_y	.27	.268	.268	.024	.024	.024	.14	.139	.139
$\sigma(Exposure_k)$ (%)	4.875	4.873	4.873	4.875	4.873	4.873	4.875	4.873	4.873
Adjusted R^2	0.602	0.600	0.601	0.479	0.482	0.483	0.582	0.580	0.580
County-Level FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Household Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Regional Controls	No	No	Yes	No	No	Yes	No	No	Yes

Notes: The table reports the results of the estimation of our difference-in-differences specification described in Equation (1) where the outcome variables are given by the (standardised) scores of each party at time t calculated as in Equation (5) using the bag of words for populist rhetoric and salience of bank-related topics obtained using the seededLDA algorithm on the representatives' parliamentary speeches for each party, and matched to the individual political preferences from the survey data. The variable of interest is the (standardised) exposure to the credit shock of county k, as calculated in Equation (4), interacted with an indicator variable equal to one for all years after the occurrence of the credit shock. Both outcome and treatment are expressed in standard deviations. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.

policy stances. These results are particularly interesting when compared with the estimates presented in Columns 7-9. In this set of columns, the dependent variable captures the intention to vote for parties that use populist rhetoric *and* talk frequently about the topic of banking and financial crisis. Also in this case, the coefficient of the credit shock is positive and significant at the one per cent level. This result indicates that the credit shock had a positive effect on individual intentions to vote for populist parties that focused on banking and financial crisis. However, it should be noted that the coefficient of the shock is largest when the dependent variable is only populism (Columns 4-6), also when compared to the combined dependent variable (Columns 7-9). This means that the credit shock mostly rewarded parties that adopted a populist rhetoric, holding constant their focus on the topic of banking and financial crisis. Table 5 provides the results based on the dictionary approach using parliamentary speeches. The estimates are not significantly different from the baseline results.

8 Political Preferences as a Reaction to the Local Economic Distress

In the previous section we identified the overall effect of the credit shock on populist preferences. However, a key challenge when disentangling the effect is to understand what the main drivers are that link the credit supply shock to the revealed political preferences. One limitation of this paper is related to the firm-level data and the missed opportunity to link individuals with their employers if they are employed. In an ideal framework, we would observe the direct effect of the credit shock on individuals employed by firms that had relationships with Commerzbank before the shock, and thus of the potential experience of a firm-level credit contraction on employees. Nevertheless, a direct link would not enable us to explain the channel for the joint reaction of both sides on labour market demand. Instead, we focus on capturing the perceived economic insecurity of individ-

Table 5: The Effect of the Credit Shock on Political Preferences: Outcome as Dictionary Scores

	Banking and Financial Crisis			**Populism**			**Combined**		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$Exposure_k \times Post_t$	0.037***	0.039***	0.043***	0.136***	0.135***	0.143***	0.052***	0.054***	0.058***
	(0.006)	(0.006)	(0.006)	(0.022)	(0.024)	(0.024)	(0.008)	(0.008)	(0.008)
Number of Observations	161,680	154,562	154,562	161,680	154,562	154,562	161,680	154,562	154,562
Number of Counties	400	400	400	400	400	400	400	400	400
$\bar{y}$	.84	.842	.842	.121	.121	.121	.48	.482	.482
σ_y	.247	.245	.245	.029	.029	.029	.126	.126	.126
$\sigma(Exposure_k)$ (%)	4.875	4.873	4.873	4.875	4.873	4.873	4.875	4.873	4.873
Adjusted R^2	0.900	0.902	0.902	0.435	0.439	0.440	0.872	0.874	0.875
County-Level FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Household Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Regional Controls	No	No	Yes	No	No	Yes	No	No	Yes

Notes: The table shows the results of the estimation of our difference-in-differences specification described in Equation (1) where the outcome variables are given by the (standardised) scores of each party at time t calculated as in Equation (5) using the bag of words for populist rhetoric and salience of bank-related topics that we use as seeds included as priors in the seededLDA algorithm on the representatives' parliamentary speeches for each party, and matched to the individual political preferences from the survey data. The variable of interest is the (standardised) exposure to the credit shock of county k, as calculated in Equation (4), interacted with an indicator variable equal to one for all years after the occurrence of the credit shock. Both outcome and treatment are expressed in standard deviations. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.

uals located in areas where the credit shock hit hardest. Individuals in more exposed counties perceive a widespread economic insecurity after the shock takes place, independently of their direct link to the shock *per se*.[26] This perceived insecurity maps onto three channels identified in the political economy literature as drivers of populist demand: deteriorating expectations about the local economy, declining trust in mainstream political institutions (not just the incumbent), and frustration directed at the financial sector as the proximate cause of distress (Guriev and Papaioannou, 2022; Cerqua, Ferrante and Letta, 2023). Against this backdrop, we expect individuals to be more likely to radicalise and express dissatisfaction with the political system and punish mainstream parties. This explains the shift in demand for populism. We test this hypothesis by analysing, at county level, whether areas that are more affected by the credit shock experience a contraction in economic outcomes compared to counties that are less affected beforehand. In practice, we estimate the following specification:

$$y_{kt} = \delta_k + \lambda_t + \sum_{r \neq 2008} \beta_r \left[\mathbf{1} \left(Exposure_k > s \right) \times \mathbf{1} \left(t = r \right) \right] + \gamma \boldsymbol{X}_k + \eta_{kt} \tag{7}$$

where y_{kt} denotes the local economic indicator of interest for county k, δ_k and λ_t are county and year fixed effects, respectively, $Exposure_k$ is the exposure to the credit shock measured as dependence on Commerzbank of county k as calculated in Equation (4), s indicates a preferred cut-off for the exposure distribution and $\boldsymbol{X}_k$ are time-invariant covariates at county level. The set of coefficients β_r capture the year-by-year effect of the credit shock in county k on the specific economic indicator y_{kt} relative to the first lag.

Figure 12 plots the estimates from equation (7) for the natural logarithm of the county GDP for each given year using the doubly robust difference-in-differences estimator in Sant'Anna and Zhao (2020), and adopting the median as the cut-off for the treatment

[26] Huber (2018) provides evidence that the credit shock has no effect on household credit, given features of the German financial system that facilitate switching banks. At most, aggregate lending to commercial banks rose from 2007 to 2010, showing that other commercial banks were able to re-balance the consequences of Commerzbank's lending cut on households.

variable in analogy to Figure 9. Each dot indicates the estimate of the interaction between the year indicator and the treatment above the median indicator variable, with bars indicating 95% confidence intervals. These estimates show semi-elasticity between counties where the pre-shock exposure to the credit shock is higher than the median of the treatment distribution and counties up to the threshold on the logarithm of the local GDP, re-weighting the outcome evolution for the covariates. We observe that, prior to 2008, the difference in GDP was not significant and constant over time, whereas from 2009 onwards there is a rapid decline in the difference in GDP for counties that are more affected by the shock, except for the first lead which is not significant. The ATT measured from the leads of the estimation in Figure 12 indicates a decrease in regional GDP of around 2.5% for counties that are exposed to the credit shock above the median after the shock takes place. In a similar specification with two-way fixed effects, where the continuous treatment interacts with an indicator variable equal to one for the years after the shock, we find an ATT of -2.1% in local GDP for a one standard deviation increase in exposure to the credit shock. These estimates are comparable to the evidence in Huber (2018), which finds an average GDP loss of 2.2 percent from a standard deviation increase in exposure to the credit shock. Huber (2018) documents extensively that the temporary lending cut propagated from firms to the real economy and may have contributed to the sluggish recovery from the Great Recession even though the banking sector had stabilised by 2010, without any sign of convergence to the level of less affected counties in the subsequent two years.[27] Figure 12 not only confirms this evidence, but also suggests that the conver-

[27] Huber (2018) contributes to an emerging literature that studies the effect of lending cuts on the economy. Using a selected sample of firms, he finds a negative direct real effect on firms dependent on Commerzbank following the credit shock, which reduced their bank debt, capital stock and employment, and which propagates into the real economy with spillover or indirect effects that impact producers of non-tradables and firms with high innovation activities independently of their direct connections to Commerzbank when the aggregate economic environment of a county responds to the lending cut. Amiti and Weinstein (2018) decompose aggregate loan movements in Japan for the period 1990-2010 into bank, firm and industry common shocks, finding that supply-side financial shocks have a large impact on firms' investments. Similarly to Huber (2018), Chodorow-Reich (2014) studies the effect of bank lending frictions on unemployment outcomes, finding that firms that had pre-crisis relationships with less healthy lenders had a lower likelihood of obtaining a loan following the Lehman Brothers bankruptcy in the United States and that they reduced employment by more than pre-crisis clients of healthier lenders. Bentolila, Jansen and Jiménez (2018) show

gence might not yet have happened.

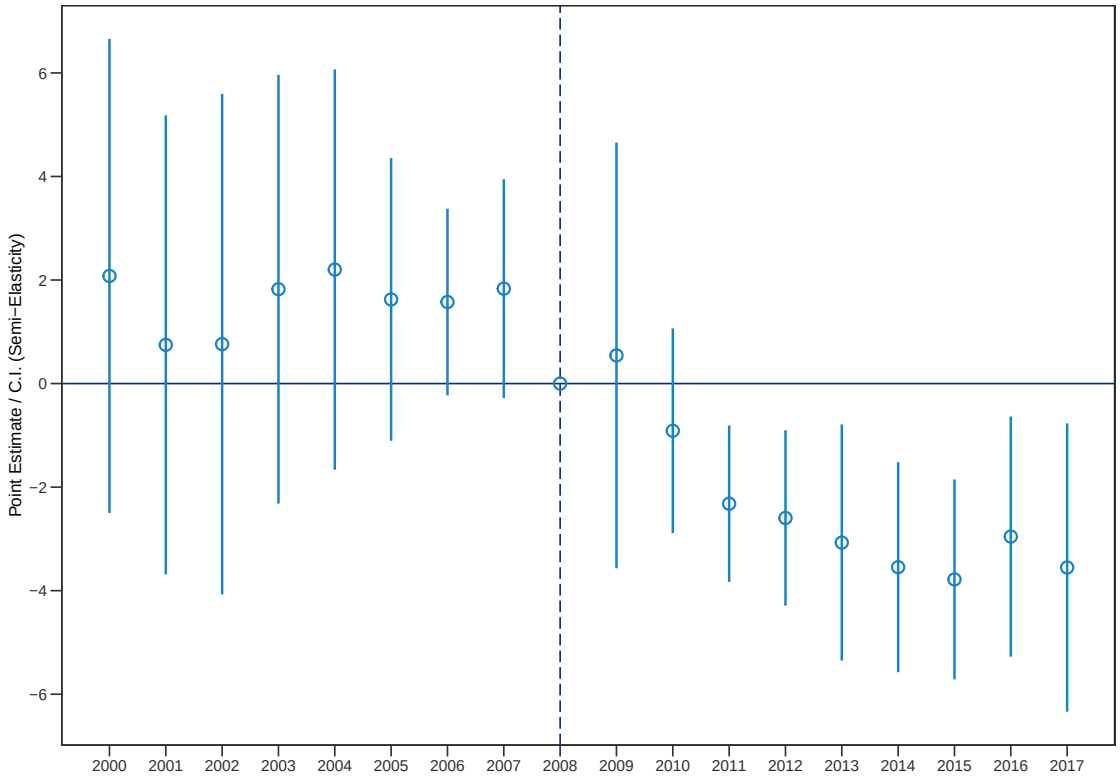
In Figure **??**, we show estimates from equation (7) for the natural logarithm of county-level employment per 1,000 units using the same method as in Figure 12. From the pre-shock point estimates and the few significant coefficients in the leads after the shock, we cannot exclude the existence of a pre-trend on employment, even though the ATT measured from the leads of the estimation indicates a significant decrease in employment of around 0.9% for counties that are exposed to the credit shock above the median after the shock takes place. Similarly, using the continuous treatment interacted with the post dummy in a similar specification with two-way fixed effects, we find an ATT of -0.75% in local employment for a one standard deviation increase in exposure to the credit shock. As suggested in Burda (2011), a combination of factors that provided disincentives for employers to lay off workers during the economic downturn may explain the meagre effect on employment.[28]

To sum up, the evidence suggests that the credit shock generated general economic distress at the local level that perseveres long after the credit shock and, in those counties that experienced more severe economic insecurity, individuals are more likely to shift their political preferences towards more populist parties. However, it does not seem to be the case that the effect is only driven by a specific category of voters who are more directly affected by the credit shock, what we would call an ego-tropic prediction of electoral preferences. In this case, citizens preoccupied with their wallets would support candidates and parties that advance their own economic interests and oppose candidates

that, during the Great Recession in Spain, a credit shock was responsible for 24% of job losses in client firms of weak banks compared to comparable firms with no significant pre-crisis relationship with weak banks. Finally, Gutierrez, Jaume and Tobal (2023) study the effect of positive bank credit supply shocks available to small and medium firms on fostering formal employment in Mexico, finding a positive and sizeable effect compared to studies in more developed countries.

[28] Burda (2011) only identifies overall cuts in hours per worker during the Great Recession, due to the reticence of employers to hire in the preceding expansion, wage moderation and the adoption of working time accounts. In a separate analysis, we test for the effect of the credit shock on furloughing (*kurzarbeitergeld*), accounting for the total months of this short-term measure in 2009 and 2010 at the individual level, but find no significant effect when comparing individuals who experienced this outcome with those who did not. The results for unemployment insurance are similar.

Figure 12: The Effect of the Credit Shock on Local Economic Performance: Difference-in-Differences Estimates



Notes: In this graph, we describe the evolution of the semi-elasticity of the local GDP between treated and untreated counties after the occurrence of the credit shock. A county k is considered as treated when its exposure to the credit shock – calculated as in equation (4) with firms given equal weight in the county and excluding finance and public sector firms from the calculation – lies above a cut-off s of the treatment distribution. In this figure, we qualify the cut-off as $s = med\,(Exposure_k)$, which is the median of the treatment distribution. Regressions are estimated at county level on a balanced panel of 401 counties from 2000 to 2017 using the doubly-robust difference-in-differences estimator of Sant'Anna and Zhao (2020), which uses weighted least squares to estimate the outcome regressions and inverse probability tilting to estimate the propensity score. The included covariates that serve as re-weighting are an indicator variable for East Germany and an indicator variable for counties exposed to a similar simultaneous crisis of the *Landesbanken* (Puri, Rocholl and Steffen, 2011). Year regression coefficients of interest from equation (7) are interactions between an indicator variable equal to one for treated counties and year fixed effects and are estimated relative to the omitted interaction with the first lag before the occurrence of the credit shock. Coefficient estimates of the year interactions are plotted as dots with their 95% confidence intervals indicated by vertical lines. All the point estimates and confidence bands are re-scaled by 100 to be interpreted as semi-elasticities from the baseline, and standard errors are clustered and bootstrapped at county level. The weighted average of all time average treatment effects on the treated is -2.47% (std. err. 0.0082). Estimating the coefficient of interest β for the continuous treatment in the following specification with two-way fixed effects:

$$y_{kt} = \delta_k + \lambda_t + \beta Exposure_k \times Post_t + \boldsymbol{X}_{kt} \times Post_t + \iota_{kt}$$

where $\boldsymbol{X}_{kt}$ includes a dummy for East Germany, a dummy for Landesbanken in Crisis (Puri, Rocholl and Steffen, 2011), the logarithm of population, population density and GDP per capita at county level, we obtain a 2.1% decrease in GDP for each standard deviation increase in the treatment dose ($std.err. = 0.0059$, $t = -3.551$).

who appear to threaten them (Kinder and Kiewiet, 1981). Rather, the political preference response is most influenced by the overall local economic condition and the sense of general increased insecurity. This is mostly explained as a socio-tropic reaction compared to a more pocketbook reaction. The socio-tropic citizens are going to vote more according to the country's pocketbook, not their own, which is reflected in the perceived economic situation in their local territory. Those voters will move towards parties that promise to be a contrast to the mainstream that placed people's economic well-being in jeopardy. There is a recurrent argument in political science (Kinder and Kiewiet, 1981; Lewis-Beck and Stegmaier, 2000; Lewis-Beck, Nadeau and Elias, 2008; Colantone and Stanig, 2018) that asserts that voters punish mainstream parties when the economy is weak in favour of parties that promise to restore economic well-being and fight against the elites that caused distress – *i.e.* the banks and the more traditional establishment that supported them, as we discussed in the previous section. We provide further evidence in the analysis in Figure **??**. It plots estimates of the coefficient of the interaction between exposure to the credit shock and various moderators in a regression with our binary indicator of individual populist preferences as the outcome variable and using a balanced panel of individuals from 2006 to 2012, including time, county and individual fixed effects. The point estimates express the differential effect of the indicated pre-shock sub-population over the baseline effect of the credit shock on individual populist preferences after the occurrence of the shock. Despite finding negative point estimates for retired, house-owning and higher-educated individuals and positive estimates for the other considered mediating factors, only the additive effect on unemployed individuals is slightly significant with 10%, supporting our thesis.

Are there some residual pre-shock differences between counties? In Table 6, we try to address this question. Using our preferred specification of equation (1) with county and year fixed effects, and individual, household and regional controls, we estimate the effect of the credit shock on political preferences, selecting individuals in county sub-samples

based on pre-shock and time-invariant characteristics. In Columns 2 and 3, we restrict estimates to solely rural or urban areas, respectively, using the German denomination in the name of the county as an indicator. We see that the credit shock has a higher magnitude and significant effect on populist preferences of individuals based in rural areas compared with individuals residing in cities.[29] We then analyse pre-shock differences in GDP growth and employment, measured as the average annual growth rate between 2001 and 2006. Splitting the individual sample into counties above and below the 25th percentile for each measure, we find significant differences in the effect of different average pre-shock GDP growth rates on the response to the credit shock in terms of populist preferences, we find a slightly higher effect for counties where employment growth was slow during the last economic expansion. However, we should be careful when interpreting these results as they might highlight unobserved differences between those counties that we are not able to capture with this specification. Instead, we would need to fully saturate the regression, which would yield noisy results.

9 Drivers of the Populist Preferences Response

In the previous section, we provided evidence that the credit shock has real consequences for the local economy that have translated into individuals changing their political preferences towards populist parties focusing on their county's finances. However, it is possible that there are multiple margins along which heterogeneity in populist preferences may arise even considering a generalised shock. It might be that the general average movement towards populist parties generated by the credit shock identified by the ATT estimated from equation (1), includes considerable differences in the response that are driven by particular features of those individuals. In this section, we address this question by taking a machine learning approach to revealing the key dimensions of the het-

[29]This is in line with Ziblatt, Hilbig and Bischof (2024), who explore the reason why those who vote for far-right parties in Germany tend to cluster in rural communities.

Table 6: The Differential Effect of the Credit Shock on Political Preferences: County Characteristics

	Full Sample	Urban Areas	Rural Areas	$\overline{\Delta GDP_k}^{2001-2006}$		$\overline{\Delta Employment_k}^{2001-2006}$	
				$\geq 25^{th}$	$< 25^{th}$	$\geq 25^{th}$	$< 25^{th}$
$Exposure_k \times Post_t$	0.549***	0.287	0.424**	0.395**	0.901***	0.470**	1.060***
	(0.190)	(0.351)	(0.196)	(0.177)	(0.287)	(0.238)	(0.386)
Number of Observations	362,122	108,637	253,485	275,308	86,814	274,095	88,027
Number of Counties	401	107	294	302	99	302	99
Outcome Mean (%)	3.373	3.944	3.109	3.053	4.374	2.481	6.684
$\sigma(Exposure_k)$ (%)	4.861	4.405	3.965	4.742	5.061	5.102	3.613
Adjusted R^2	0.059	0.059	0.059	0.053	0.076	0.052	0.055
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	No	No	No	No	No	No	No
Individual Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Regional Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table explores the difference in the effect of the credit shock on populist preferences among different county-level, pre-shock or time-invariant characteristics. It shows the results of the estimation of our difference-in-differences preferred specification, analogue to Column 4 of Table 1, excluding controls for GDP and GDP per capita. Column 1 shows the estimate of the effect of the credit shock on the full sample. Columns 2 and 3 report results of the coefficient of splitting urban and rural areas, respectively. In Columns 4 and 5 we split the sample, considering the average annual growth rate between 2001 and 2006 for counties above and below the 25th percentile, respectively. Finally, in Columns 6 and 7 we split the sample considering the average annual growth in employment between 2001 and 2006 for counties above and below the 25th percentile, respectively. For all sub-samples, we include county and year fixed effects, individual, household and regional controls for variables like population density and share of foreigners. The table includes the number of observations, the number of counties, the outcome mean and the standard deviation unit of the treatment for all the results. The coefficients of interest are scaled by 100 to be interpreted as the percentage point increase in the outcome mean of one-σ units increase of the treatment after the occurrence of the shock. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at 10%, 5%, and 1% levels, respectively. In Table **??** we provide additional evidence of different splits in the distribution of average annual growth and employment growth, and looking at the two growth rates in 2006.

erogeneity. For this purpose, we first need to determine the individualised treatment effects deriving from our general difference-in-differences settings. In this exercise, for each individual receiving a certain exposure to the credit shock (dose) d determined by the county of residence, we calculate her individualised treatment effect comparing her with a counterfactual individual constructed as the average individual receiving any other dose, after averaging the individual outcomes for the periods before and after the shock, respectively.[30] In practice, we map individuals i into a group g given their assigned dose $d \in Exposure(k)$, and we match each individual to a synthetic individual assigned to a dose $d' = \mathbb{E}\left[Exposure_k | \, d' \neq d\right]$, which is the leave-out mean of all the doses other than the dose d received by i. Then, we define a two-period specification $\tau \in \{0, 1\}$ averaging over the outcome variable $\mathbf{1}\left(Populist = 1\right)_{it}$ for the periods before and after the credit shock for both each individual i and their assigned synthetic individual, who will be assigned to the average outcome of all individuals treated with a different dose than individual i for the two periods. Hence, for each group g, we estimate the following equation:

$$y_{i\tau} = \alpha_g + \delta \cdot D_i + \lambda \cdot \mathbf{1}\left(\tau = 1\right) + \beta_i \cdot D_i \times \mathbf{1}\left(\tau = 1\right) + \gamma \boldsymbol{X}_i + \zeta_{i\tau} \tag{8}$$

where $y_{i\tau}$ measures the average outcome of interest for individual i in the period τ, D_i identifies the treatment received by individual i – either the group dose or the average leave-out dose, $\boldsymbol{X}_i$ is a vector of characteristics that we may want to take into account and $\zeta_{i\tau}$ captures other determinants of the populist preferences. $\hat{\beta}_i$ are the estimates of causal effects at the individual level given by the difference-in-differences specification in Equation (8).[31] Next, we consider the total response in populist preferences as a function

[30] This means that, for each individual, we must have at least one period before and after the shock, which is not guaranteed by the rolling panel applied in Table 1, and this reduces the total sample size. When averaging over time for the periods before and after the shock, we calculate a weighted average using the sampling weights for those variables that are affected by sampling at the individual level.

[31] The weighted average of the individualised treatment effects that we obtain through this exercise should be equal to the ATT identified by Equation (1) with the correct weights.

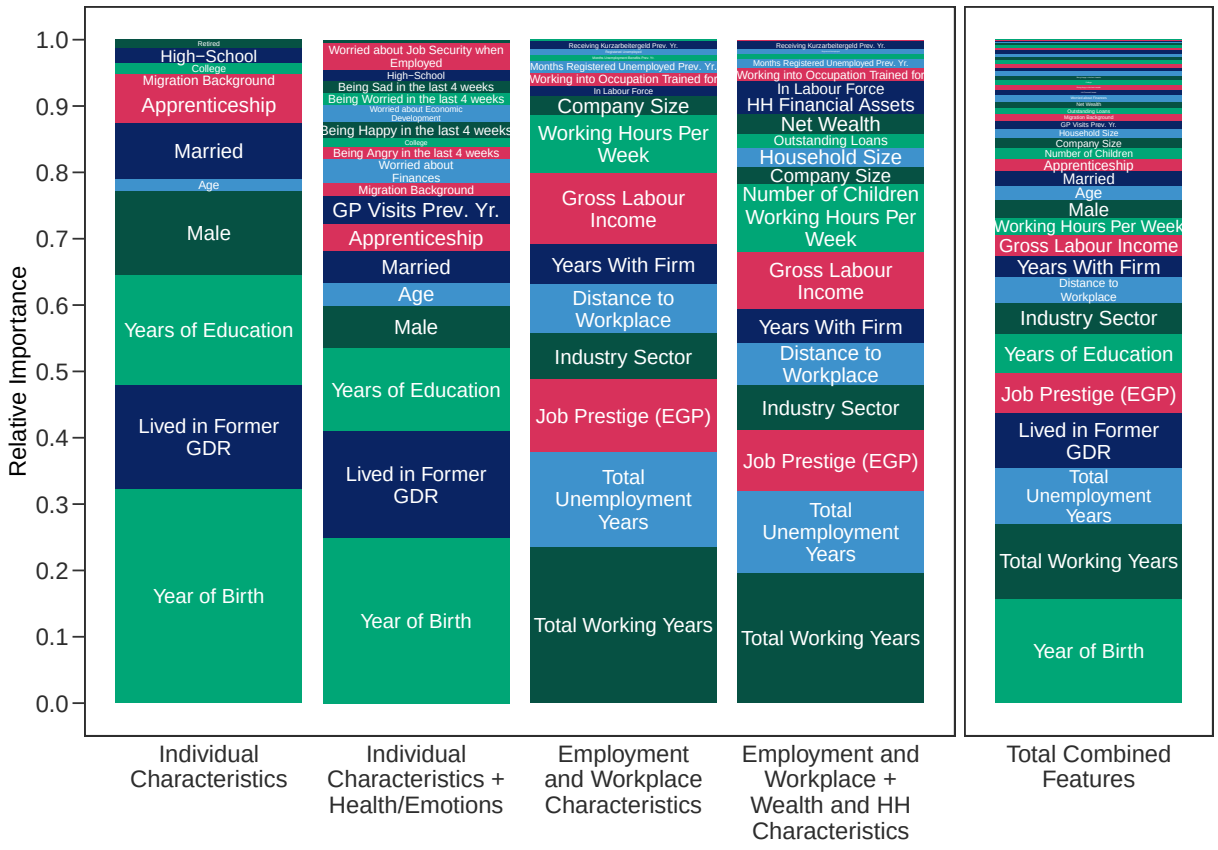
of the individualised treatment effects of a matrix of individual characteristics that are shaping the individual response in populist preferences, $\beta \rightarrow \beta_i(\boldsymbol{X}_{it})$, using an extreme gradient-boosted tree (XGB) algorithm (Chen and Guestrin, 2016) to predict the response in populist preferences using $\boldsymbol{X}_{it}$ defined as a sparse set of individual characteristics (features vector). XGB is a highly effective and efficient supervised learning algorithm that has been shown to provide state-of-the-art results in many classification tasks, independently of the dichotomous or continuous nature of the assigned target variable. It is particularly well known for being robust to outliers and for automatically handling feature selection. Building on the general gradient tree boosting method (Friedman, 2001), this is preferable to the classic random forest algorithm (Breiman, 2001) because of its training process, leaving the output interpretation unchanged but building predictions on the previously trained decision trees.[32]

In Figure 13, we look at the relative informativeness of each feature in predicting the response in populist preferences to the credit shock at the individual level, measuring the prediction gain achieved by the gradient-boosted tree splitting along the dimension of a given feature, *i.e.* its importance relative to all the other features. We estimate four different models including different and/or progressively added features, in addition to a model including all the available features to gain an overview of what features contribute most to explaining the causal effects.[33]

[32]Other methods exploit the random forest algorithm in Breiman (2001) to identify heterogeneous treatment effects using a target variable at the level of observational unit averaging numerous causal trees, where trees differ from one another due to subsampling (Athey and Imbens, 2016, 2022). The estimation of conditional average treatment effects (CATE) with *causal forest* is not preferable for two reasons. First, the methodology in Athey and Imbens (2016) and Athey and Imbens (2022) implies random assignment, which is naturally achieved by randomisation. In our case, we develop a difference-in-differences design as in Equation (1), which relies on the (strong) common trends assumption to achieve orthogonality. Therefore, using causal forests, the unconfoundedness assumption would decay if we did not extract the individualised treatment effects *ex ante* as in Equation (8) as the target variable of supervised learning. Second, the ATT in Equation (1) arises from a county-level treatment, which would not allow us to decompose until we allocate the individual contribution to the county's populist preference response of each individual due to the credit shock. For an overview of supervised machine learning methods in economics, see Athey and Imbens (2019).

[33]Here, we provide a summary of the characteristics included in the different models. Individual characteristics include birth year, age, residence in East Germany before Reunification, marital status, gender, migrant status, years of education, indicators for high-school, college and apprenticeship, retirement sta-

Figure 13: Feature Importance in Explaining Populist Preferences Response



Notes: This figure describes the relative informativeness of each feature in predicting the response in populist preferences triggered by the credit supply shock. We measure this by first estimating the causal effects at individual level in a two-period difference-in-differences setting averaging outcomes before and after the shock, where we compare each individual with an assigned dose d with the average individual receiving a different dose d' that we rationalise to the leave-out average dose excluding d. Contrary to what we specify in equation (8), we include no controls in the estimation of the individualised treatment effects. In Figure **??** we show negligible differences in gains when including controls for the estimation of individualised treatment effects. We use the estimates of the causal effects at the individual level as the target variable in an extreme gradient-boosted tree algorithm (Chen and Guestrin, 2016) where we predict the response in populist preferences using features at individual-year level matched with the individual level estimates. The importance of each feature is given by the gain in prediction achieved over all trees by splitting along the dimension of a given feature. We normalise the total gain to one, giving the relative importance of each feature in each model. In the prediction, we first proceed through cross-validation, using the entire sample, to identify the best iteration of trees (settings: rounds = 25000, sample folds = 10, early stopping rounds = 25, metric = RMSE, objective function = squared error regression). Then, we predict the target variable using the best iteration achieved through the cross-validation. Details of the features included in the different models can be found in section 9.

In the first bar of Figure 13, a model based only on individual characteristics highlights three main important features that explain the response in populist preferences: East Germany heritage, year of birth and education. In the second bar, we add proxies for health and emotions to the individual characteristics, which seem to play a minor role compared to the three main features. In the third and fourth bars, we show a model based on employment and workplace characteristics and an additive model including wealth and household characteristics, respectively. A key finding arises from these two models: cumulative employment status and occupation prestige play a major role in explaining the response in populist preferences. This also applies in the last bar, where we show a model including all the previously used features. The main takeaway from this evidence is that, even though the response to the credit supply is widespread through the perception of the general economic difficulties in the local economy, the response is driven by those features that characterise individuals as more or less 'vulnerable'. This is similar to the 'outsider status' in Dal Bó et al. (2022): individuals who are already at the margin of the socio-economic status might recognise themselves more in the rhetoric of populist parties when they propose an alternative to the establishment, characterising those parties as the alternative to the 'elite' that endangered their local economy.[34]

tus, and living in urban areas. Health and emotions include the number of GP visits, worries about own finances, job security if employed and general economic development, being sad, worried, happy and angry in the last four weeks. Employment and workplace characteristics include being in the labour force, registered as unemployed, total years of unemployment, total working years, months of unemployment and unemployment insurance in the previous year, gross labour income, job prestige scale (EGP), being furloughed, weeks on furlough in the previous year, number of contractual working hours, industry sector, company size, seniority within the firm, commuting distance, and working in the job for which one was trained for. Wealth and household characteristics include individual net wealth, household size, number of children in the household, household disposable income, hand-to-mouth indicator (Kaplan, Violante and Weidner, 2014), household financial assets, household net wealth and presence of outstanding loans. Total combined features are all the previous features taken together.

[34]In Figure **??**, we plot the summary plots of the Shapley values from the combined model with features decreasing in the value of their mean SHAP score (Lundberg and Lee, 2017; Lundberg et al., 2020). In terms of robustness, the ranking provided by the average Shapley values is similar to the relative importance plot using the simple gains from the gradient-boosted trees.

10 Conclusion

Banking crises carry not only economic consequences, but political ones too. In this paper, we provide causal evidence of this by exploiting an exogenous lending shock affecting German counties during the 2007–08 financial crisis. We show that individuals living in more exposed areas became significantly more likely to support populist parties: a one standard deviation increase in exposure raises support by around 0.5 percentage points. This corresponds to roughly a 16 percent increase relative to the baseline level of populist preferences in our survey data.

The mechanism we uncover is socio-tropic rather than direct. Voters start supporting populist parties as a reaction to the persistent deterioration of local economic conditions rather than to changes in their own personal budget. The prolonged, local stagnation generated by the credit shock fuels anti-establishment sentiment, favouring parties that combined populist rhetoric with explicit attention to banking and financial issues. Labour market history further mediates individual responses: those persistently marginalised in the labour market are especially reactive, consistent with a broader pattern in which economic outsiders are most susceptible to populist mobilisation.

As with any study exploiting a single country-specific shock, there is a natural tension between internal and external validity. The specificity of our setting is a key strength of this paper's empirical design. The clean exogenous variation provided by the Commerzbank shock allows us to isolate a causal effect that cross-country studies, which typically face more severe identification challenges, cannot easily establish. Whether the mechanisms we document generalise to other institutional contexts, party systems, or types of financial disruption nonetheless remains an open empirical question that we leave to future research.

Understanding the causes of populism is particularly important, especially since a growing body of research has exposed its potentially significant economic and social consequences. Funke, Schularick and Trebesch (2023), for instance, estimate that populist

governments reduce GDP per capita and consumption by around 10 percent in the long run, with the effects primarily driven by right-wing populists in recent decades and by left-wing populists before the 1990s. Beyond its macroeconomic costs, populism has been shown to erode institutional quality (Scheurer, Sturm and de Haan, 2026), bureaucratic competence (Bellodi, Morelli and Vannoni, 2024), and to fuel hate crimes and social polarisation (Bursztyn, Egorov and Fiorin, 2020; Romarri, 2020; Müller and Schwarz, 2021, 2023). As a result, investigating the determinants of populist support is not only of academic interest, but also central for evaluating the long-term stability and quality of democratic and economic institutions. Our findings contribute to this agenda by establishing that disruptions in credit markets are a key part of that story.

Several avenues remain open for future research. First, the text-based measure of populism we develop here could be extended and applied comparatively across other parliaments and episodes of financial distress, providing a richer picture of how the supply side of populism responds to financial and economic shocks. Second, our results raise the question of whether the political consequences of credit shocks are symmetric. If credit contractions fuel populism, do credit expansions reduce it? Testing this hypothesis by exploiting episodes of exogenous credit supply expansions would provide a more complete picture of how credit markets shape the demand for populism over the political cycle. Third, while our heterogeneity analysis points to labour market history as the key individual predictor of populist response, it leaves open why some persistently marginalised workers mobilise politically while others stop expressing any party preferences. This leads to the broader question on whether some individuals are more prone to translating the collective perception of local economic decline identified in this paper into populist voting. Combining the findings in this paper with evidence from experimental economics and political psychology could greatly benefit our knowledge on this issue.

References

Ahlquist, John, Mark Copelovitch, and Stefanie Walter. 2020. "The Political Consequences of External Economic Shocks: Evidence from Poland." *American Journal of Political Science*, 64(4): 904–920.

Alabrese, Eleonora, Sascha O. Becker, Thiemo Fetzer, and Dennis Novy. 2019. "Who Voted for Brexit? Individual and Regional Data Combined." *European Journal of Political Economy*, 56: 132–150.

Algan, Yann, Sergei Guriev, Elias Papaioannou, and Evgenia Passari. 2017. "The European Trust Crisis and the Rise of Populism." *Brookings Papers on Economic Activity*, 309–382.

Amiti, Mary, and David E. Weinstein. 2018. "How Much Do Idiosyncratic Bank Shocks Affect Investment? Evidence from Matched Bank-Firm Loan Data." *Journal of Political Economy*, 126(2): 525–587.

Anelli, Massimo, Italo Colantone, and Piero Stanig. 2019. "We Were the Robots: Automation and Voting Behavior in Western Europe." IZA Institute of Labor Economics Discussion Paper 12485, 2019-07.

Angrist, Joshua D., and Guido W. Imbens. 1995. "Two-Stage Least Squares Estimation of Average Causal Effects in Models with Variable Treatment Intensity." *Journal of the American Statistical Association*, 90(430): 431–442.

Antoniades, Alexis, and Charles W. Calomiris. 2020. "Mortgage Market Credit Conditions and U.S. Presidential Elections." *European Journal of Political Economy*, 64: 101909.

Arzheimer, Kai. 2015. "The AfD: Finally a Successful Right-Wing Populist Eurosceptic Party for Germany?" *West European Politics*, 38(3): 535–556.

Arzheimer, Kai, and Carl C. Berning. 2019. "How the Alternative for Germany (AfD) and Their Voters Veered to the Radical Right, 2013–2017." *Electoral Studies*, 60: 102040.

Ash, Elliott, and Stephen Hansen. 2023. "Text Algorithms in Economics." *Annual Review of Economics*, 15(1): 659–688.

Athey, Susan, and Guido W. Imbens. 2016. "Recursive Partitioning for Heterogeneous Causal Effects." *Proceedings of the National Academy of Sciences*, 113(27): 7353–7360.

Athey, Susan, and Guido W. Imbens. 2019. "Machine Learning Methods Economists Should Know About." *Annual Review of Economics*, 11: 685–725.

Athey, Susan, and Guido W. Imbens. 2022. "Design-Based Analysis in Difference-In-Differences Settings with Staggered Adoption." *Journal of Econometrics*, 226(1): 62–79.

Autor, David, David Dorn, Gordon Hanson, and Kaveh Majlesi. 2020. "Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure." *The American Economic Review*, 110(10): 3139–3183.

Avdeenko, Alexandra. 2018. "Long-Term Evidence of Retrospective Voting: A Natural Experiment from the German Democratic Republic." *European Economic Review*, 103: 83–107.

Baker, Scott R., Nicholas Bloom, and Steven J. Davis. 2016. "Measuring Economic Policy Uncertainty." *The Quarterly Journal of Economics*, 131(4): 1593–1636.

Barros, Laura, and Manuel Santos Silva. 2019. "#EleNão: Economic Crisis, the Political Gender Gap, and the Election of Bolsonaro." Ibero-America Institute for Economic Research (IAI) Working Paper 242, Göttingen, Germany.

Becker, Irene, and Richard Hauser, ed. 2000. *The Personal Distribution of Income in an International Perspective.* Berlin, Germany:Springer.

Bellodi, Luca, Massimo Morelli, and Matia Vannoni. 2024. "A Costly Commitment: Populism, Economic Performance, and the Quality of Bureaucracy." *American Journal of Political Science*, 68(1): 193–209.

Bentolila, Samuel, Marcel Jansen, and Gabriel Jiménez. 2018. "When Credit Dries Up: Job Losses in the Great Recession." *Journal of the European Economic Association*, 16(3): 650–695.

Berg, Tobias, Markus Reisinger, and Daniel Streitz. 2021. "Spillover Effects in Empirical Corporate Finance." *Journal of Financial Economics*, 142(3): 1109–1127.

Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-In-Differences Estimates?" *The Quarterly Journal of Economics*, 119(1): 249–275.

Blei, David M., Andrew Y. Ng, and Michael I. Jordan. 2003. "Latent Dirichlet Allocation." *Journal of Machine Learning Research*, 3: 30.

Braggion, Fabio, Alberto Manconi, and Haikun Zhu. 2020. "Credit and Social Unrest: Evidence from 1930s China." *Journal of Financial Economics*, 138(2): 295–315.

Breiman, Leo. 2001. "Random Forests." *Machine Learning*, 45(1): 5–32.

Breyer, Magdalena. 2022. "Populist Positions in Party Competition: Do Parties Strategically Vary Their Degree of Populism in Reaction to Vote and Office Loss?" *Party Politics*, 13540688221097082.

Burda, Jennifer Hunt and Michael C. 2011. "What Explains the German Labor Market Miracle in the Great Recession? The Evolution of Inflation Dynamics and the Great Recession." *Brookings Papers on Economic Activity*.

Bursztyn, Leonardo, Georgy Egorov, and Stefano Fiorin. 2020. "From Extreme to Mainstream: The Erosion of Social Norms." *The American Economic Review*, 110(11): 3522–3548.

Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna. 2025. "Difference-in-Differences with a Continuous Treatment."

Campante, Filipe, Ruben Durante, and Francesco Sobbrio. 2018. "Politics 2.0: The Multifaceted Effect of Broadband Internet on Political Participation." *Journal of the European Economic Association*, 16(4): 1094–1136.

Cantarella, Michele, Nicolò Fraccaroli, and Roberto Volpe. 2023. "Does fake news affect voting behaviour?" *Research Policy*, 52(1): 104628.

Cantoni, Davide, Felix Hagemeister, and Mark Westcott. 2020. "Persistence and Activation of Right-Wing Political Ideology." Working Paper.

Cerqua, Augusto, Chiara Ferrante, and Marco Letta. 2023. "Electoral Earthquake: Local Shocks and Authoritarian Voting." *European Economic Review*, 156: 104464.

Ceyhan, Sara. 2016. "Konservativ oder doch schon rechtspopulistisch? Die politischen Positionen der AfD-Parlamentskandidaten im Parteienvergleich." *Zeitschrift für Politikwissenschaft*, 26(1): 49–76.

Chen, Tianqi, and Carlos Guestrin. 2016. "XGBoost: A Scalable Tree Boosting System." 785–794. San Francisco California USA:ACM.

Chodorow-Reich, Gabriel. 2014. "The Employment Effects of Credit Market Disruptions: Firm-level Evidence from the 2008–9 Financial Crisis." *The Quarterly Journal of Economics*, 129(1): 1–59.

Coffé, Hilde, and Rebecca Plassa. 2010. "Party Policy Position of Die Linke: A Continuation of the PDS?" *Party Politics*, 16(6): 721–735.

Colantone, Italo, and Piero Stanig. 2018. "The Trade Origins of Economic Nationalism: Import Competition and Voting Behavior in Western Europe." *American Journal of Political Science*, 62(4): 936–953.

Dal Bó, Ernesto, Frederico Finan, Olle Folke, Torsten Persson, and Johanna Rickne. 2022. "Economic and Social Outsiders but Political Insiders: Sweden's Populist Radical Right." *The Review of Economic Studies*, rdac037.

de Bromhead, Alan, Barry Eichengreen, and Kevin O'Rourke. 2012. "Right-Wing Political Extremism in the Great Depression." National Bureau of Economic Research Working Paper 17871, Cambridge, MA.

Degryse, Hans, and Steven Ongena. 2005. "Distance, Lending Relationships, and Competition." *The Journal of Finance*, 60(1): 36.

Dehdari, Sirus H. 2022. "Economic Distress and Support for Radical Right Parties—Evidence From Sweden." *Comparative Political Studies*, 55(2): 191–221.

Dinas, Elias, Konstantinos Matakos, Dimitrios Xefteris, and Dominik Hangartner. 2019. "Waking up the Golden Dawn: Does Exposure to the Refugee Crisis Increase Support for Extreme-Right Parties?" *Political Analysis*, 27(2): 244–254.

Dippel, Christian, Robert Gold, Stephan Heblich, and Rodrigo Pinto. 2022. "The Effect of Trade on Workers and Voters." *The Economic Journal*, 132(641): 199–217.

Doerr, Nicole. 2021. "The Visual Politics of the Alternative for Germany (AfD): Anti-Islam, Ethno-Nationalism, and Gendered Images." *Social Sciences*, 10(1): 20.

Doerr, Sebastian, Stefan Gissler, José-Luis Peydró, and Hans-Joachim Voth. 2022. "Financial Crises and Political Radicalization: How Failing Banks Paved Hitler's Path to Power." *The Journal of Finance*.

Ellger, Fabio, Hanno Hilbig, Sascha Riaz, and Philipp Tillmann. 2024. "Local Newspaper Decline and Political Polarization – Evidence from a Multi-Party Setting." *British Journal of Political Science*, 54(4): 1256–1275.

Enke, Benjamin. 2020. "Moral Values and Voting." *Journal of Political Economy*, 128(10): 3679–3729.

Enke, Benjamin, Ricardo Rodríguez-Padilla, and Florian Zimmermann. 2023. "Moral Universalism and the Structure of Ideology." *The Review of Economic Studies*, 90(4): 1934–1962.

Ferner, Cornelia, Clemens Havas, Elisabeth Birnbacher, Stefan Wegenkittl, and Bernd Resch. 2020. "Automated Seeded Latent Dirichlet Allocation for Social Media Based Event Detection and Mapping." *Information-an International Interdisciplinary Journal*, 11(8): 376.

Ferrara, Federico Maria. 2022. "Why Does Import Competition Favor Republicans? Localized Trade Shocks and Cultural Backlash in the US." *Review of International Political Economy*, 0(0): 1–24.

Fetzer, Thiemo. 2019. "Did Austerity Cause Brexit?" *The American Economic Review*, 109(11): 3849–3886.

Fetzer, Thiemo, Srinjoy Sen, and Pedro CL Souza. 2020. "Housing Insecurity, Homelessness and Populism: Evidence from the UK." Centre for Economic Policy Research Discussion Paper 14184, London, UK.

Freeden, Michael. 1998. "Is Nationalism a Distinct Ideology?" *Political Studies*, 46(4): 748–765.

Frey, Carl Benedikt, Thor Berger, and Chinchih Chen. 2018. "Political Machinery: Did Robots Swing the 2016 US Presidential Election?" *Oxford Review of Economic Policy*, 34(3): 418–442.

Friedman, Jerome H. 2001. "Greedy Function Approximation: A Gradient Boosting Machine." *Annals of Statistics*, 29(5): 1189–1232.

Funke, Manuel, Moritz Schularick, and Christoph Trebesch. 2016. "Going to Extremes: Politics After Financial Crises, 1870–2014." *European Economic Review*, 88: 227–260.

Funke, Manuel, Moritz Schularick, and Christoph Trebesch. 2023. "Populist Leaders and the Economy." *American Economic Review*, 113(12): 3249–3288.

Galofré-Vilà, Gregori, Christopher M. Meissner, Martin McKee, and David Stuckler. 2021. "Austerity and the Rise of the Nazi Party." *Journal of Economic History*, 81(1): 81–113.

Garro, Haritz. 2021. "Political Consequences of Economic Hardship: State Economic Activity and Polarization in American Legislatures." *The Journal of Law, Economics, and Organization*, 37(3): 512–533.

Gennaro, Gloria, Giampaolo Lecce, and Massimo Morelli. 2024. "Intertemporal Evidence on the Strategy of Populism in the United States." *Electoral Studies*, 92: 102853.

Gethin, Amory, Clara Martínez-Toledano, and Thomas Piketty. 2022. "Brahmin Left Versus Merchant Right: Changing Political Cleavages in 21 Western Democracies, 1948–2020." *The Quarterly Journal of Economics*, 137(1): 1–48.

Giannetti, Mariassunta, and Steven Ongena. 2012. ""Lending by Example": Direct and Indirect Effects of Foreign Banks in Emerging Markets." *Journal of International Economics*, 86(1): 167–180.

Giuliano, Paola, and Romain Wacziarg. 2020. "Who Voted for Trump? Populism and Social Capital." National Bureau of Economic Research Working Paper 27651, NBER Working Paper Series.

Goebel, Jan, Markus M. Grabka, Stefan Liebig, Martin Kroh, David Richter, Carsten Schröder, and Jürgen Schupp. 2019. "The German Socio-Economic Panel (SOEP)." *Jahrbücher für Nationalökonomie und Statistik*, 239(2): 345–360.

Grabka, Markus M., and Jan Goebel. 2018. "Income Distribution in Germany: Real Income on the Rise Since 1991 but More People with Low Incomes." DIW Berlin Report 21/2018, Berlin, Germany.

Guiso, Luigi, Helios Herrera, Massimo Morelli, and Tommaso Sonno. 2019. "Global Crises and Populism: The Role of Eurozone Institutions." *Economic Policy*, 34(97): 95–139.

Guiso, Luigi, Helios Herrera, Massimo Morelli, and Tommaso Sonno. 2024. "Economic Insecurity and the Demand for Populism in Europe." *Economica*, 91(362): 588–620.

Guiso, Luigi, Paola Sapienza, and Luigi Zingales. 2004. "Does Local Financial Development Matter?" *The Quarterly Journal of Economics*, 119(3): 929–969.

Guriev, Sergei, and Elias Papaioannou. 2022. "The Political Economy of Populism." *Journal of Economic Literature*, 60(3): 753–832.

Gutierrez, Emilio, David Jaume, and Martín Tobal. 2023. "Do Credit Supply Shocks Affect Employment in Middle-Income Countries?" *American Economic Journal: Economic Policy*, 15(4): 1–36.

Gyöngyösi, Győző, and Emil Verner. 2022. "Financial Crisis, Creditor-Debtor Conflict, and Populism." *The Journal of Finance*, 77(4): 2471–2523.

Hassan, Tarek A., Stephan Hollander, Laurence van Lent, and Ahmed Tahoun. 2019. "Firm-Level Political Risk: Measurement and Effects." *The Quarterly Journal of Economics*, 134(4): 2135–2202.

Herrera, Helios, Guillermo Ordoñez, and Christoph Trebesch. 2020. "Political Booms, Financial Crises." *Journal of Political Economy*, 128(2): 507–543.

Hobolt, Sara B., and Catherine de Vries. 2016. "Turning Against the Union? The Impact of the Crisis on the Eurosceptic Vote in the 2014 European Parliament Elections." *Electoral Studies*, 44: 504–514.

Huber, Kilian. 2018. "Disentangling the Effects of a Banking Crisis: Evidence from German Firms and Counties." *The American Economic Review*, 108(3): 868–898.

Ivashina, Victoria, and David Scharfstein. 2010. "Bank Lending During the Financial Crisis of 2008." *Journal of Financial Economics*, 97: 319–338.

Jankowski, Michael, Sebastian Schneider, and Markus Tepe. 2017. "Ideological Alternative? Analyzing Alternative Für Deutschland Candidates' Ideal Points via Black Box Scaling." *Party Politics*, 23(6): 704–716.

Kalemli-Ozcan, Sebnem, Bent Sorensen, Carolina Villegas-Sanchez, Vadym Volosovych, and Sevcan Yesiltas. 2015. "How to Construct Nationally Representative Firm Level Data from the Orbis Global Database: New Facts and Aggregate Implications." National Bureau of Economic Research Working Paper 21558, Cambridge, MA.

Kalemli-Özcan, Şebnem, Luc Laeven, and David Moreno. 2022. "Debt Overhang, Rollover Risk, and Corporate Investment: Evidence from the European Crisis." *Journal of the European Economic Association*, 20(6): 2353–2395.

Kaplan, Greg, Giovanni L. Violante, and Justin Weidner. 2014. "The Wealthy-Hand-to-Mouth." Brookings Institution BPEA Article, Washington, DC.

Kinder, Donald R., and D. Roderick Kiewiet. 1981. "Sociotropic Politics: The American Case." *British Journal of Political Science*, 11(2): 129–161.

Klapper, Leora, Luc Laeven, and Raghuram Rajan. 2006. "Entry Regulation as a Barrier to Entrepreneurship." *Journal of Financial Economics*, 82(3): 591–629.

Klein, Sebastian. 1993. *Strategien der Großbanken in den neuen Bundesländern. Trends in Finance and Banking*, Gabler Verlag.

Lewis-Beck, Michael S., and Mary Stegmaier. 2000. "Economic Determinants of Electoral Outcomes." *Annual Review of Political Science*, 3(1): 183–219.

Lewis-Beck, Michael S., Richard Nadeau, and Angelo Elias. 2008. "Economics, Party, and the Vote: Causality Issues and Panel Data." *American Journal of Political Science*, 52(1): 84–95.

Lichter, Andreas, Max Löffler, and Sebastian Siegloch. 2020. "The Long-Term Costs of Government Surveillance: Insights from Stasi Spying in East Germany." *Journal of the European Economic Association*.

Lu, Bin, Myle Ott, Claire Cardie, and Benjamin K. Tsou. 2011. "Multi-Aspect Sentiment Analysis with Topic Models." 81–88. Vancouver, BC, Canada.

Lundberg, Scott M., and Su-In Lee. 2017. "A Unified Approach to Interpreting Model Predictions." Vol. 30. Long Beach, CA:Curran Associates, Inc.

Lundberg, Scott M., Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. 2020. "From Local Explanations to Global Understanding with Explainable AI for Trees." *Nature Machine Intelligence*, 2(1): 56–67.

Mudde, Cas. 2004. "The Populist Zeitgeist." *Government and Opposition*, 39(4): 541–563.

Mukand, Sharun, and Dani Rodrik. 2018. "The Political Economy of Ideas: On Ideas Versus Interests in Policymaking." National Bureau of Economic Research Working Paper 24467, Cambridge, MA.

Müller, Karsten, and Carlo Schwarz. 2021. "Fanning the Flames of Hate: Social Media and Hate Crime." *Journal of the European Economic Association*, 19(4): 2131–2167.

Müller, Karsten, and Carlo Schwarz. 2023. "From Hashtag to Hate Crime: Twitter and Antiminority Sentiment." *American Economic Journal: Applied Economics*, 15(3): 270–312.

Norris, Pippa. 2020. "Measuring Populism Worldwide." *Party Politics*, 26(6): 697–717.

Norris, Pippa, and Ronald Inglehart. 2019. *Cultural Backlash: Trump, Brexit, and Authoritarian Populism.* Cambridge, UK:Cambridge University Press.

Panunzi, Fausto, Nicola Pavoni, and Guido Tabellini. 2024. "Economic Shocks and Populism." *The Economic Journal*, 134(663): 3047–3061.

Puri, Manju, Jörg Rocholl, and Sascha Steffen. 2011. "Global Retail Lending in the Aftermath of the US Financial Crisis: Distinguishing Between Supply and Demand Effects." *Journal of Financial Economics*, 100(3): 556–578.

Rauh, Christian, and Jan Schwalbach. 2020. "The Parlspeech V2 Data Set: Full-Text Corpora of 6.3 Million Parliamentary Speeches in the Key Legislative Chambers of Nine Representative Democracies."

Romarri, Alessio. 2020. "Do Far-Right Mayors Increase the Probability of Hate Crimes? Evidence From Italy." Job Market Paper.

Rooduijn, Matthijs, and Teun Pauwels. 2011. "Measuring Populism: Comparing Two Methods of Content Analysis." *West European Politics*, 34(6): 1272–1283.

Rooduijn, Matthijs, Stijn Van Kessel, Caterina Froio, Andrea Pirro, Sarah De Lange, Daphne Halikiopoulou, Paul Lewis, Cas Mudde, and Paul Taggart. 2019. "The PopuList: An Overview of Populist, Far Right, Far Left and Eurosceptic Parties in Europe."

Sant'Anna, Pedro H.C., and Jun Zhao. 2020. "Doubly Robust Difference-in-Differences Estimators." *Journal of Econometrics*, 219(1): 101–122.

Sartre, Emilie, Gianmarco Daniele, and Paul Vertier. 2020. "Toxic Loans and the Rise of Populist Candidacies." Working Paper.

Scheurer, Aline, Jan-Egbert Sturm, and Jakob de Haan. 2026. "Does Populism Erode Central Bank Independence?" *Open Economies Review*.

Schmitt-Beck, Rüdiger. 2017. "The 'Alternative Für Deutschland in the Electorate': Between Single-Issue and Right-Wing Populist Party." *German Politics*, 26(1): 124–148.

Shapiro, Adam Hale, Moritz Sudhof, and Daniel J. Wilson. 2020. "Measuring News Sentiment." *Journal of Econometrics*.

Steinmayr, Andreas. 2021. "Contact versus Exposure: Refugee Presence and Voting for the Far Right." *The Review of Economics and Statistics*, 103(2): 310–327.

van der Loo, Mark P. J. 2014. "The Stringdist Package for Approximate String Matching." *The R Journal*, 6(1): 111–122.

Watanabe, Kohei, and Yuan Zhou. 2020. "Theory-Driven Analysis of Large Corpora: Semisupervised Topic Classification of the UN Speeches." *Social Science Computer Review*.

Winkler, William. 1990. "String Comparator Metrics and Enhanced Decision Rules in the Fellegi-Sunter Model of Record Linkage." *Proceedings of the Section on Survey Research Methods*.

Zhuravskaya, Ekaterina, Maria Petrova, and Ruben Enikolopov. 2020. "Political Effects of the Internet and Social Media." *Annual Review of Economics*, 12(1): 415–438.

Ziblatt, Daniel, Hanno Hilbig, and Daniel Bischof. 2024. "Wealth of Tongues: Why Peripheral Regions Vote for the Radical Right in Germany." *American Political Science Review*, 118(3): 1480–1496.